

Statistical models as natural transformations: meaningfulness, coherence and priors as states in Markov categories

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Abstract

We show that a statistical model in the sense of McCullagh, in the form given by Brøns, is a natural transformation between two functors from the category of designs to the Kleisli category Stoch of the Giry monad, provided that its components are measurable in the parameter. The condition is empty for finite models. A design-indexed quantity is a family of morphisms of Stoch defined on the parameter objects, called meaningful if it is natural. We prove that Tjur’s criterion, imposed on parameter functions indexed by finite samples with multiplicities, forces the indexing by the support and then coincides with naturality over the insertions. For finite designs we show that a quantity can be corrected to a natural one within a given class of corrections if and only if a class vanishes in the first cohomology group of a Baues–Wirsching complex relative to that class, while its image in the absolute group is always zero. In the one-way layout the marginal dispersion is not meaningful, and the within-group dispersion is its unique correction which leaves the merged design unchanged. A prior is a family of states on the parameter objects, called coherent over a class of design morphisms if it is natural over that class. We show that coherence at a merge confines the prior to the image of the corresponding parameter map, that coherence over the insertions is Kolmogorov consistency, and that coherence over the injections adds the exchangeability assumed by the categorical de Finetti theorem. In the finite one-way scheme the coherent priors form polytopes of known dimension. The analogue of Jeffreys’ general rule is not coherent, while the analogue of his rule for location–scale families is. Finally we show that ridge regression is the Bayesian inversion of the Gaussian linear model with respect to a Gaussian prior, which is coherent over the insertions and never over the injections.

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Keywords: statistical model; Markov category; Giry monad; natural transformation; meaningfulness; design category; Baues–Wirsching cohomology; Bayesian inversion; coherence; exchangeability; Jeffreys prior; ridge regression.

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1 Introduction

Let S be a measurable space and let $\mathcal{P}(S)$ be the set of probability measures on S . A statistical model is usually defined as a map $\Theta_0 \rightarrow \mathcal{P}(S)$, where Θ_0 is a set, i.e. as a parametrised family of probability measures on a fixed sample space. McCullagh observed in [21] that this definition does not say how the family must transform when the experimental design changes. Therefore it cannot state the requirement that a parameter retain its meaning across designs. His remedy was categorical. Designs form a category, sample spaces and parameter spaces are functors on it, and a model is a natural transformation.

In his discussion of [21], Brøns [5, pp. 1280–1282] organised these data into a hexagonal diagram of categories and functors. He observed that the resulting notion of model is equivalent, by a general argument on comma categories, to a diagram in a fixed “category of statistical models” (see Proposition 2.5). He also noted that this category is too small, since to include all statistical transformations “Markov kernels must be allowed” [5, p. 1282]. This remark was the starting point of the present work.

In the same volume, Tjur [26] considered the rules which assign a parameter, i.e. a function of the unknown distribution, to each finite sample of covariate values, repetitions allowed. He called such a rule a *parameter function*, and he called it *meaningful* when it is invariant under the formation of marginal models, i.e. when reducing the design by removing some of the sampled x ’s leaves the associated parameter unchanged. His non-example is the design-averaged expectation $\alpha + \beta\bar{x}$. The criterion of McCullagh [21, §4.5] is that a subparameter be a natural transformation of functors on the design category, and in his rejoinder he reads Tjur’s remarks as explaining what is meant by a natural parameter in regression [21, p. 1304]. The two criteria are stated over different classes of morphisms, namely design reduction alone for Tjur and the full design category for McCullagh, and neither source asserts that they coincide.

In [21] the codomain of the construction is the category **Set** of sets, where conditioning, sufficiency and almost sure equality cannot be expressed. Markov categories [10, 7, 23, 11] provide a categorical probability theory in which they can. The aim of this paper is to replace **Set** with the Markov category **Stoch**, i.e. the Kleisli category of the Giry monad, whose objects are the measurable spaces and whose morphisms are the Markov kernels, and to study the resulting notion of model.

Let $\mathbf{D}$ be the category of designs and let $\mathbf{V}$ be the category of response scales (see Definition 2.1). A statistical model in the sense of McCullagh and Brøns is a natural transformation $P: L \Rightarrow R$ between two functors $L, R: \mathbf{V} \times \mathbf{D}^{\text{op}} \rightarrow \mathbf{Set}$, where L gives the parameter sets and R gives the sets of the probability measures on the sample spaces (see Definition 2.2). We denote by $\delta: \mathbf{Meas} \rightarrow \mathbf{Stoch}$ the functor which is the identity on the objects and maps a measurable map f to the kernel $x \mapsto \delta_{f(x)}$. Suppose that the parameter sets are endowed with measurable structures such that the reparametrisation maps are measurable. Composing with δ we obtain two functors $\mathbf{L}, \mathbf{R}: \mathbf{V} \times \mathbf{D}^{\text{op}} \rightarrow \mathbf{Stoch}$. We shall prove the following.

Theorem 1.1 (Bridge). *There is a canonical bijection between the statistical models $P: L \Rightarrow R$ whose components are measurable in the parameter and the natural transformations $\mathbf{P}: \mathbf{L} \Rightarrow \mathbf{R}$. The morphisms $\mathbf{L}(m)$ and $\mathbf{R}(m)$ are deterministic for every morphism m of $\mathbf{V} \times \mathbf{D}^{\text{op}}$. If the design data are finite, then every statistical model has measurable components, and **Stoch** can be replaced by its full subcategory **FinStoch** of the finite discrete spaces, whose morphisms are the stochastic matrices.*

Remark 1.2. Theorem 1.1 is proved in Section 4, see Theorem 4.1 and Theorem 4.4. The proof is a transposition under the bijection $\mathbf{Stoch}(X, Y) \cong \mathbf{Meas}(X, \mathcal{G}Y)$. The measurability of the components is a restriction with respect to [21], where no measurability in the parameter is imposed. The role of Theorem 1.1 is to place models, quantities and priors in the same category, where the results below are stated. For the statistician this means that marginalisation, conditioning, sufficiency and Bayesian inversion become operations of the same category in which the model is a morphism.

We call $\mathbf{P}$ the *bridged model* associated to P . Our main results concern two kinds of families of morphisms of **Stoch** indexed by the covariate spaces, which are the objects of a category Ω .

Let $\Theta_{\mathbf{Meas}}(\omega)$ be the parameter object at the covariate space ω and let $\Lambda: \Omega^{\text{op}} \rightarrow \mathbf{Meas}$ be a functor. A *design-indexed quantity* is a family of morphisms $g_\omega: \Theta_{\mathbf{Meas}}(\omega) \rightsquigarrow \Lambda(\omega)$ of **Stoch**. It is called *coherent* over a subcategory $\mathbf{Q}$ of Ω if it is natural over $\mathbf{Q}$ (Definition 5.1), *meaningful* if it is coherent over Ω , and *Tjur-natural* if it is coherent over the subcategory Ω_{red} generated by the proper inclusions of covariate spaces, which we call *insertions* after Tjur. The parameter functions of Tjur are indexed by the finite samples of covariate values, with multiplicities, and not by the covariate spaces. Accordingly, a *sample-indexed quantity* is a family of morphisms $G_d: \Theta_{\mathbf{Meas}}(\omega) \rightsquigarrow \Lambda(\omega)$ indexed by the designs $d = (u, \omega, x)$ with x surjective. A *Tjur reduction* of d is the morphism of designs given by the removal of some units together with the covariate values which are no longer observed, and G is called *Tjur-meaningful* if (12) holds at every Tjur reduction (Definition 5.7).

Theorem 1.3 (Tjur's criterion). *Let $\Omega = \Omega_{\text{lab}}$ be the category of the finite non-empty sets and let $\mathbf{U} = \mathbf{FinSet}_{\text{inj}}$ be the category of the finite sets and injective maps, as in the one-way layout, with an arbitrary parameter functor. A sample-indexed quantity G is Tjur-meaningful if and only if $G_{(u, \omega, x)} = g_\omega$ depends only on the support ω of the sample and the design-indexed quantity g is Tjur-natural. In this case $g_\omega = G_{(\omega, \omega, \text{id}_\omega)}$. Every meaningful quantity is Tjur-natural, and the converse does not hold in general.*

Remark 1.4. Theorem 1.3 is proved in Section 5, see Proposition 5.9 for the general hypotheses. Thus Tjur's criterion first forces the indexing by the support and then coincides with naturality over the insertions, which is weaker than the criterion of McCullagh. The non-examples of Tjur are sample-indexed and not support-indexed. The quantity of Example 5.13 is natural at a morphism ψ if and only if ψ is injective. The regression examples, whose covariate spaces are vector spaces, are not covered by Theorem 1.3.

Suppose now that Ω is a finite category and that the sets $\Theta(\omega)$ and $\Lambda(\omega)$ are finite. Let $k \supseteq \mathbb{Q}$ be a field and let $A(\omega) = k^{\Theta(\omega)}$ and $B(\omega) = k^{\Lambda(\omega)}$ be the algebras of functions. Let $C^\bullet$ be the complex of Baues and Wirsching [2] of Ω with coefficients in $\text{Hom}_k(B, A)$, so that $H^0 = \text{Nat}(B, A)$. A deterministic quantity g gives the 0-cochain $g^* = (g_\omega^*: B(\omega) \rightarrow A(\omega))$, and g is meaningful if and only if its *defect* $c_g = d^0 g^*$ is

zero. For a linear subspace $C \subseteq C^0$ we denote by $C_C^\bullet$ the subcomplex of $C^\bullet$ with $C_C^0 = C$ and $C_C^n = C^n$ for $n \geq 1$.

Theorem 1.5 (Obstruction). *There is $h \in C$ such that $g^* + h$ is natural if and only if the class of c_g in $H^1(C_C^\bullet)$ is zero, and h is unique modulo $C \cap \text{Nat}(B, A)$. The image of this class in $H^1(\Omega; \text{Hom}_k(B, A))$ is zero for every g . In the one-way layout with two groups and finitely many values of the mean and of the dispersion, let C_1 be the subspace of the 0-cochains which vanish at the merged design. Then $C_1 \cap \text{Nat}(B, A) = 0$, and the marginal dispersion has a unique correction in C_1 , which gives the within-group dispersion.*

Remark 1.6. Theorem 1.5 is proved in Section 6, see Propositions 6.4, 6.6 and 6.8. Thus the absolute group H^1 is an invariant of the design scheme and not an obstruction group for quantities. We compute it in Example 6.12, where for a constant target it is given by copies of the first derived limit of the parameter diagram. When the mean takes at least two values, the marginal dispersion is not meaningful, and it has no injective natural recalibration (Propositions 6.9 and 6.11).

A prior for a bridged model is a family of states $\pi_\omega: I \rightsquigarrow \Theta_{\text{Meas}}(\omega)$, i.e. of probability measures on the parameter objects, and it is coherent over $\mathbf{Q}$ when $\pi_\omega = \Theta_{\text{Meas}}(\psi)_* \pi_{\omega'}$ for all $\psi: \omega \rightarrow \omega'$ in $\mathbf{Q}$ (Definition 8.1). Thus quantities transform contravariantly with the parameter, and priors covariantly. Let Ω_{inj} be the subcategory of Ω generated by the insertions and the isomorphisms.

Theorem 1.7 (Priors). *Let π be a prior for a bridged model.*

1. *If π is coherent at $\psi: \omega \rightarrow \omega'$, then π_ω vanishes on the measurable sets disjoint from the image of $\Theta_{\text{Meas}}(\psi)$.*
2. *If π is coherent over Ω_{inj} , then π_ω is invariant under $\Theta_{\text{Meas}}(\text{Aut } \omega)$ for every ω . Coherence over Ω_{red} imposes no such symmetry.*
3. *Let $\mathbf{P}$ be the Gaussian linear model with known variance σ_0^2 , and let $\pi_\omega^\tau = \mathcal{N}(0, \tau^2 I_\omega) \otimes \delta_{\sigma_0^2}$ be the prior associated to a choice of inner products on the covariate spaces which restrict along the insertions, where I_ω is the induced form on ω^* . Then the Bayesian inversion of $\mathbf{P}_d$ with respect to π_ω^τ exists in BorelStoch , and it is the Gaussian kernel whose mean is the ridge estimator with penalty $\lambda = \sigma_0^2/\tau^2$. The prior π^τ is coherent over Ω_{red} , and it is coherent at a linear automorphism A if and only if A is orthogonal. In particular, it is not coherent over Ω_{inj} .*

Remark 1.8. Theorem 1.7 is proved in Sections 8 and 9, see Propositions 8.2, 9.2 and 9.4. By part (1), coherence at a merge of covariate values confines the prior to the image of the corresponding parameter map, which is a diagonal in the one-way layout. For this reason naturality of priors is required over Ω_{red} or Ω_{inj} and not over Ω . Coherence over Ω_{red} is Kolmogorov consistency, and the invariance of part (2) is the exchangeability assumed by the categorical de Finetti theorem of Fritz, Gonda and Perrone [11] (Remark 8.7). Part (3) corresponds to the usual instruction to standardise the covariates. In the finite one-way scheme, the coherent priors form polytopes whose dimensions we compute (Proposition 8.4), and the analogue of Jeffreys' general rule is not coherent, while the analogue of his rule for location–scale families is (Proposition 8.5). Improper priors and the marginalisation paradoxes of Dawid, Stone and Zidek [9], which cannot arise from proper priors, require a codomain with unnormalised states, and they are left open (Remark 8.6).

The closest work is the thesis of Patterson [22], an algebraic theory of statistical models with machine representation in view, where the signature of a model is primitive, while here the structure indexed by the designs is. Fritz [10] proves synthetic theorems on sufficiency inside the hom-sets of one Markov category [10, Def. 14.1], while the naturality of McCullagh and Brøns is a condition across a diagram of such hom-sets. Conditionals, almost sure equality and Bayesian inversion are taken from [10] and from Cho and Jacobs [7] (see Section 7). The Kolmogorov products of Rischel and Fritz [24] and the categorical de Finetti theorem are used only in Remark 8.7, and we do not apply them to the designs of Definition 2.1. Information cohomology [1, 28] has coefficients and indexing categories different from those of Section 6, and we do not construct a comparison map. Jeffreys' general rule [17], his recommendation for location–scale families and his objection to the general rule in the problem of several means [18] are used as they

are presented by Kass and Wasserman [19] and by Consonni, Fouskakis, Liseo and Ntzoufras [8]. Proposition 8.5 agrees with the preference for reference priors [4, 3] over the multidimensional Jeffreys rule, but we do not claim that coherence characterises them. Ridge regression is due to Hoerl and Kennard [15], and its reading as a Gaussian posterior mean is classical. What is new in Theorem 1.7(3) is its description as a Bayesian inversion indexed by the designs, together with the coherence of its prior.

The paper goes as follows. In Section 2 we recall the framework of McCullagh and Brøns, with a complete proof of the equivalence stated by Brøns (Proposition 2.5), and in Section 3 the notions on Markov categories that we need. Sections 4, 5 and 6 concern models and quantities, and we refer to their setting as the *Fisherian shape*. Sections 7, 8 and 9 concern priors, and we refer to their setting as the *Bayesian shape*. In Appendix A we identify the category of statistical models of Brøns, on the objects which satisfy two regularity hypotheses, with the arrows of *Stoch* whose domain is a copower of the monoidal unit (Proposition A.4), and we show that it is a coreflective subcategory of its measurable analogue (Proposition A.5). The rest of the paper does not depend on the appendix.

2 The McCullagh–Brøns framework

We recall the definition of a statistical model given by McCullagh in [21], in the form given by Brøns in [5]. All the categories considered in this section are locally small.

2.1 Statistical models

Definition 2.1. The *design data* consist of the following.

1. A category $\mathbf{U}$, whose objects are called *statistical units*.
2. A category $\mathbf{\Omega}$, whose objects are called *covariate spaces*.
3. A category $\mathbf{V}$, whose objects are called *response scales*.
4. The category $\mathbf{S} := \mathbf{V} \times \mathbf{U}^{\text{op}}$ of *sample space indices* and a functor $\Gamma: \mathbf{S} \rightarrow \mathbf{Meas}$. The measurable space $\Gamma(v, u)$ is called the *sample space* of the v -valued responses on the units u .
5. Two functors $J_{\mathbf{U}}: \mathbf{U} \rightarrow \mathbf{C}$ and $J_{\mathbf{\Omega}}: \mathbf{\Omega} \rightarrow \mathbf{C}$ into a category $\mathbf{C}$. The comma category

$$\mathbf{D} := (J_{\mathbf{U}} \downarrow J_{\mathbf{\Omega}})$$

is called the category of *designs*. Its objects are the triples (u, ω, x) , where $x: J_{\mathbf{U}}u \rightarrow J_{\mathbf{\Omega}}\omega$ is a morphism of $\mathbf{C}$ called the *design map*. A morphism $(u, \omega, x) \rightarrow (u', \omega', x')$ is a pair (φ, ψ) , where $\varphi: u \rightarrow u'$ is a morphism of $\mathbf{U}$ and $\psi: \omega \rightarrow \omega'$ is a morphism of $\mathbf{\Omega}$ such that $x' \circ J_{\mathbf{U}}\varphi = J_{\mathbf{\Omega}}\psi \circ x$. We denote by $\text{pr}_{\mathbf{U}}: \mathbf{D} \rightarrow \mathbf{U}$ and $\text{pr}_{\mathbf{\Omega}}: \mathbf{D} \rightarrow \mathbf{\Omega}$ the projections.

6. A category $\mathbf{K}$ of *parameter objects* with a faithful functor $U_{\mathbf{K}}: \mathbf{K} \rightarrow \mathbf{Set}$, and a functor $\Theta: \mathbf{\Omega}^{\text{op}} \rightarrow \mathbf{K}$ called the *parameter functor*.

In the examples, the objects of $\mathbf{U}$ are finite sets of experimental units and its morphisms are injective maps, i.e. inclusions of subsamples. The morphisms of $\mathbf{V}$ are measurable maps between response scales, such as coarsenings of the response or affine rescalings (see Remark 2.14 for the latter). Furthermore $\mathbf{C} = \mathbf{Set}$ and $J_{\mathbf{U}}, J_{\mathbf{\Omega}}$ are the underlying-set functors, so that a design map assigns a covariate value to each unit, and $\Gamma(v, u) = v^u$ with the product structure.

Let $\mathcal{P}: \mathbf{Meas} \rightarrow \mathbf{Set}$ be the functor which maps a measurable space to the set of its probability measures and a measurable map to the pushforward. It is the composition of the Giry monad $\mathcal{G}: \mathbf{Meas} \rightarrow \mathbf{Meas}$ (see [14] and Section 3) with the forgetful functor. Note that a map $\Theta_0 \rightarrow \mathcal{P}(S)$ is a family of probability measures on S indexed by the set Θ_0 , and no measurability in the index is required. This is the only difference between a model and a family of morphisms of *Stoch*. It disappears when the parameter object is discrete, and it is an additional hypothesis in general (see Theorem 4.4).

Definition 2.2. Let the design data be as in Definition 2.1 and consider the functors $L, R: \mathbf{V} \times \mathbf{D}^{\text{op}} \rightarrow \mathbf{Set}$ given by

$$L := U_K \circ \Theta \circ \text{pr}_\Omega^{\text{op}} \circ \text{pr}_2 \quad \text{and} \quad R := \mathcal{P} \circ \Gamma \circ (\text{id}_V \times \text{pr}_U^{\text{op}}),$$

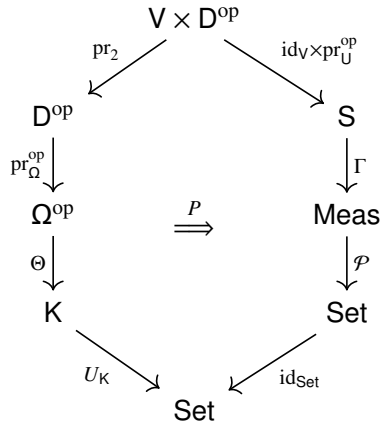
where $\text{pr}_2: \mathbf{V} \times \mathbf{D}^{\text{op}} \rightarrow \mathbf{D}^{\text{op}}$ is the projection. A *statistical model* in the sense of McCullagh and Brøns, or *McCullagh–Brøns model*, is a natural transformation

$$P: L \Rightarrow R.$$

Thus a statistical model is given by a map

$$P_{v,d}: U_K \Theta(\omega) \longrightarrow \mathcal{P}(\Gamma(v, u)), \quad \theta \longmapsto P_{v,d}(\cdot \mid \theta),$$

for every response scale v and every design $d = (u, \omega, x)$, such that the square corresponding to m commutes for every morphism m of $\mathbf{V} \times \mathbf{D}^{\text{op}}$ (see (2) and (3) below). The categories and the functors introduced so far fit into the following hexagonal diagram, due to Brøns, and a statistical model is a natural transformation between the two compositions from $\mathbf{V} \times \mathbf{D}^{\text{op}}$ to $\mathbf{Set}$.



Remark 2.3. The product categories $\mathbf{S} = \mathbf{V} \times \mathbf{U}^{\text{op}}$ and $\mathbf{V} \times \mathbf{D}^{\text{op}}$ are due to McCullagh [21, §4.7, p. 1240] and are restated by Brøns in his own notation [5, p. 1280]. In [21, §4.2] the response scale is fixed, and for fixed v the naturality squares of Definition 2.2 are the squares of [21, p. 1236]. McCullagh takes the morphisms of $\mathbf{U}$ and $\mathbf{\Omega}$ to be injective [21, §4.1]. Brøns considers this restriction dispensable and does not impose it [5, p. 1281]. In Definition 2.1 we follow Brøns for $\mathbf{\Omega}$, and this choice fixes the class of morphisms over which naturality is quantified in Section 5. In all our examples $\mathbf{U} = \mathbf{FinSet}_{\text{inj}}$, and the injectivity of the maps between units is used in the verifications of Section 2.3.

2.2 Brøns' comma-category equivalence

Let $F: \mathbf{B} \rightarrow \mathbf{E}$ be a functor. Recall that the objects of the comma category $(\text{id}_{\mathbf{E}} \downarrow F)$ are the triples (e, b, p) , where $p: e \rightarrow Fb$ is a morphism of $\mathbf{E}$.

Definition 2.4. The *category of statistical models* is the comma category

$$\text{StatMod} := (\text{id}_{\mathbf{Set}} \downarrow \mathcal{P}).$$

Its objects are the triples (Θ_0, S, p) , where Θ_0 is a set, S is a measurable space and $p: \Theta_0 \rightarrow \mathcal{P}(S)$ is a map, i.e. a parametrised family of probability measures on S . A morphism $(\Theta_0, S, p) \rightarrow (\Theta'_0, S', p')$ is a pair (r, f) , where $r: \Theta_0 \rightarrow \Theta'_0$ is a map and $f: S \rightarrow S'$ is a measurable map such that $\mathcal{P}(f) \circ p = p' \circ r$.

We have the following result.

Proposition 2.5 (Brøns). *There is a bijection, natural in all the data, between*

1. *the statistical models $P: L \Rightarrow R$ of Definition 2.2;*

2. the functors $\widehat{P}: \mathbf{V} \times \mathbf{D}^{\text{op}} \rightarrow \mathbf{StatMod}$ whose compositions with the two projections of the comma category are $L: \mathbf{V} \times \mathbf{D}^{\text{op}} \rightarrow \mathbf{Set}$ and $\Gamma \circ (\text{id}_{\mathbf{V}} \times \text{pr}_{\mathbf{U}}^{\text{op}}): \mathbf{V} \times \mathbf{D}^{\text{op}} \rightarrow \mathbf{Meas}$.

Proof. Let $G: \mathbf{A} \rightarrow \mathbf{E}$, $H: \mathbf{A} \rightarrow \mathbf{B}$ and $F: \mathbf{B} \rightarrow \mathbf{E}$ be functors. By the universal property of the comma category, the functors $\mathbf{A} \rightarrow (\text{id}_{\mathbf{E}} \downarrow F)$ whose compositions with the two projections are G and H correspond bijectively to the natural transformations $G \Rightarrow FH$. Indeed, an assignment $a \mapsto (Ga, Ha, P_a)$ on the objects is functorial if and only if the squares of P commute, and the values on the morphisms are forced. The statement follows by taking $\mathbf{A} = \mathbf{V} \times \mathbf{D}^{\text{op}}$, $\mathbf{E} = \mathbf{Set}$, $\mathbf{B} = \mathbf{Meas}$, $G = L$, $H = \Gamma \circ (\text{id}_{\mathbf{V}} \times \text{pr}_{\mathbf{U}}^{\text{op}})$ and $F = \mathcal{P}$. The naturality in the data is a straightforward verification. $\square$

Remark 2.6. In [5, pp. 1280, 1282] the result is stated with a sketch of the proof.

2.3 Examples

We give three examples of the design data of Definition 2.1, namely the one-way layout, multiple linear regression and binary classification, and we verify that they give statistical models. We also give two families which are not statistical models. We denote by $\mathbf{FinSet}_{\text{inj}}$ the category of finite sets and injective maps, and by $\mathbf{Vect}_{\mathbb{R}}^{\text{fd}}$ the category of finite dimensional real vector spaces and linear maps.

In all the examples $\mathbf{C} = \mathbf{Set}$ and $J_{\mathbf{U}}$, $J_{\mathbf{D}}$ are the underlying-set functors. Hence an object of $\mathbf{D}$ is a triple (u, ω, x) with $x: u \rightarrow \omega$ a map, and a morphism $(u, \omega, x) \rightarrow (u', \omega', x')$ is a pair (φ, ψ) such that

$$x' \circ \varphi = \psi \circ x. \quad (1)$$

In words, the covariate value of a unit of the smaller design is carried by ψ to the covariate value of the same unit in the larger design. This is the compatibility needed to compare a model across two designs. Let $v \in \text{ob } \mathbf{V}$ and let $f = (\varphi, \psi): d = (u, \omega, x) \rightarrow d' = (u', \omega', x')$ be a morphism of $\mathbf{D}$. The square of Definition 2.2 corresponding to $(\text{id}_v, f^{\text{op}})$ is

$$\begin{array}{ccc} U_{\mathbf{K}}\Theta(\omega') & \xrightarrow{P_{v,d'}} & \mathcal{P}(\Gamma(v, u')) \\ \Theta(\psi) \downarrow & & \downarrow \mathcal{P}(\text{res}_{\varphi}) \\ U_{\mathbf{K}}\Theta(\omega) & \xrightarrow{P_{v,d}} & \mathcal{P}(\Gamma(v, u)) \end{array} \quad (2)$$

where $\text{res}_{\varphi}: v'^{\prime\prime} \rightarrow v''$, $y \mapsto y \circ \varphi$, is the restriction along φ . It says that the marginal on the subsample of the model at d' is the model at d evaluated at the pulled back parameter. Let now $\alpha: v \rightarrow v'$ be a morphism of $\mathbf{V}$ and let d be fixed. Since L does not depend on $\mathbf{V}$, the corresponding square reduces to

$$\mathcal{P}(\alpha'') \circ P_{v,d} = P_{v',d} \quad \text{on } U_{\mathbf{K}}\Theta(\omega). \quad (3)$$

We write (N-U), (N- Ω) and (N-V) for the conditions obtained from (2) and (3) by letting only φ , only ψ and only α be different from the identity, respectively.

The verifications below use only (1), the finiteness of u and the injectivity of φ . The latter is necessary. Indeed, if φ is not injective, then (2) fails in the three models below, since two units with the same image have equal responses and are not independent.

2.3.1 The one-way layout

This example will be used throughout the paper (see Example 4.7, Section 5 and Example 6.7). A single factor with finitely many levels, called *treatments*, is applied to the experimental units, and each unit receives exactly one treatment. There is no other explanatory variable. A *treatment label set* is a finite non-empty set ω whose elements are the names of the treatments. It carries no further structure, and for this reason the covariate spaces below are sets and not vector spaces as in Example 2.12. The passage between the two descriptions is dummy coding.

Example 2.7.

1. **Units.** $\mathbf{U} = \mathbf{FinSet}_{\text{inj}}$. An object u is the set of the enrolled units, and a morphism $\varphi: u \hookrightarrow u'$ is the inclusion of a subsample. We write $n = |u|$.
2. **Covariate spaces.** $\Omega = \Omega_{\text{lab}}$ is the full subcategory of $\mathbf{FinSet}$ of the finite non-empty sets, i.e. of the treatment label sets. Every morphism $\psi: \omega \rightarrow \omega'$ is a surjection followed by an injection. We use the following terminology.
 - If ψ is surjective it is called a *merging* of treatments. The fibres $\psi^{-1}(b)$, with $b \in \omega'$, are the blocks of the treatments which are identified.
 - If ψ is injective and not bijective it is called a *selection* of a proper subfamily of treatments. If ψ is a proper inclusion $\omega' \subsetneq \omega$ we call it an *insertion*, or design reduction, after Tjur [26, p. 1299]. Every injection is a bijection followed by an insertion. Tjur's criterion acts on the quantities of Section 5 through the insertions only (see Lemma 5.6).
 - If ψ is bijective it is called a *relabelling* of the treatments.
 - If ψ is constant, it collapses all the treatments to a single one.

We shall also consider some subcategories of Ω_{lab} , namely the wide subcategory Ω_{merge} of the surjections and the categories $\Omega_{\text{arr}} \subset \Omega_{\text{full}}$ with two objects of Example 6.7. Naturality over a category implies naturality over its subcategories, so that Lemma 2.8 holds for all of them.

3. **Response scales.** $\mathbf{V}$ has the single object $\mathbb{R}$, with its Borel σ -algebra, and only the identity morphism. Hence (3) is empty (see Remark 2.14 for the affine maps).
4. **Sample spaces.** $\mathbf{S} = \mathbf{V} \times \mathbf{U}^{\text{op}}$ and $\Gamma(\mathbb{R}, u) = \mathbb{R}^u$ with the product σ -algebra, i.e. the space of the response vectors $y = (y_i)_{i \in u}$. The morphism $\varphi^{\text{op}}: u' \rightarrow u$ of $\mathbf{U}^{\text{op}}$ is mapped by Γ to the restriction $\text{res}_\varphi: \mathbb{R}^{u'} \rightarrow \mathbb{R}^u$, $y \mapsto y \circ \varphi$.
5. **Designs.** An object of $\mathbf{D}$ is a triple (u, ω, x) , where $x: u \rightarrow \omega$ is the *treatment assignment*, i.e. $x(i)$ is the treatment received by the unit i . Equivalently x is the partition of u into the *groups*

$$u_a := x^{-1}(a) \subseteq u, \quad a \in \omega, \quad n_a := |u_a|, \quad \sum_{a \in \omega} n_a = n,$$

some of which can be empty. The design is called *balanced* if all the n_a are equal, and x is surjective if and only if every treatment is observed. For a morphism (φ, ψ) condition (1) reads

$$x'(\varphi(i)) = \psi(x(i)) \quad (i \in u),$$

i.e. each retained unit receives in the larger design the image under ψ of the treatment it receives in the smaller one.

6. **Parameters.** The objects of $\mathbf{K}$ are the products $W \times \mathbb{R}_{>0}$ with $W \in \mathbf{Vect}_{\mathbb{R}}^{\text{fd}}$, its morphisms are the maps $A \times \text{id}_{\mathbb{R}_{>0}}$ with A linear, and U_K is the underlying-set functor. The parameter functor is

$$\Theta(\omega) = \mathbb{R}^\omega \times \mathbb{R}_{>0}, \quad \Theta(\psi)(\mu', \sigma^2) = (\mu' \circ \psi, \sigma^2),$$

i.e. $(\Theta(\psi)\mu')_a = \mu'_{\psi(a)}$. The map $\mu' \mapsto \mu' \circ \psi$ is linear, so that $\Theta(\psi)$ is a morphism of $\mathbf{K}$. A parameter value is a pair $\theta = (\mu, \sigma^2)$, where $\mu = (\mu_a)_{a \in \omega}$ is the vector of the *group means*, one for each treatment, and σ^2 is the common error variance, which is not changed by $\Theta(\psi)$.

For $d = (u, \omega, x)$ and $\theta = (\mu, \sigma^2) \in \mathbb{R}^\omega \times \mathbb{R}_{>0}$ we set

$$P_{\mathbb{R}, d}(\cdot \mid \mu, \sigma^2) := \bigotimes_{i \in u} \mathcal{N}(\mu_{x(i)}, \sigma^2) \quad \text{on } \mathbb{R}^u,$$

i.e. $Y_i = \mu_{x(i)} + \varepsilon_i$ with the ε_i independent and distributed as $\mathcal{N}(0, \sigma^2)$. This is the Gaussian one-way layout, where the group sizes n_a are given by the design and no constraint such as $\sum_a \mu_a = 0$ is imposed on μ .

Lemma 2.8. *The family P of Example 2.7 is a statistical model.*

Proof. Let $f = (\varphi, \psi): d \rightarrow d'$ be a morphism of $\mathbf{D}$ and let $\theta' = (\mu', \sigma^2) \in \Theta(\omega')$. We have $\Theta(\psi)\theta' = (\mu' \circ \psi, \sigma^2)$, hence

$$P_{\mathbb{R},d}(\cdot \mid \Theta(\psi)\theta') = \bigotimes_{i \in u} \mathcal{N}(\mu'_{\psi(x(i))}, \sigma^2).$$

On the other hand $P_{\mathbb{R},d'}(\cdot \mid \theta') = \bigotimes_{j \in u'} \mathcal{N}(\mu'_{x'(j)}, \sigma^2)$, and the pushforward along res_φ is the marginalisation onto the coordinates in $\varphi(u) \subseteq u'$. Since the factors are independent, it is equal to $\bigotimes_{i \in u} \mathcal{N}(\mu'_{x'(\varphi(i))}, \sigma^2)$. The two measures coincide by (1), so that (2) commutes. Condition (3) is empty because $\mathbf{V}$ is discrete. $\square$

The proof uses only the independence across the units and the fact that the law of the unit i depends on the design only through $x(i)$.

Remark 2.9. For the one-way layout, the conditions (N-U), (N- Ω) and (N-V) read as follows.

- (N-U) Let $\psi = \text{id}_\omega$ and let φ be a proper inclusion. The marginal of the layout on the subsample $\varphi(u)$ is the layout on that subsample, at the same (μ, σ^2) and with the assignment $x' \circ \varphi$. This is consistency under subsampling.
- (N- Ω) Let $\varphi = \text{id}_u$. If ψ is a merging, then $\mu' \circ \psi$ is constant on the blocks of ψ , and the square says that the model at ω evaluated at such a mean vector is the merged model at μ' . Note that naturality does not say that the merged groups have equal means. If ψ is injective, the same square is read on a subfamily of treatments. When ψ is an inclusion, this is design reduction, and by Lemma 5.6 it is the only part of (N- Ω) which is reached by Tjur's criterion. A general injection adds a relabelling of the surviving groups. In Example 6.10 the merging alone detects nothing, see part (i), and the selections detect the failure, see part (ii).
- (N-V) It is empty by Example 2.7(3).

Remark 2.10. By (1) an automorphism (φ, ψ) of a design (u, ω, x) is given by a relabelling ψ which preserves the group sizes and by a bijection φ which carries u_a onto $u_{\psi(a)}$ for all a . Naturality at (φ, id) is the within-group exchangeability of $P_{\mathbb{R},d}(\cdot \mid \theta)$, and naturality at a general (φ, ψ) is its equivariance under the relabelling of the treatments, which is available for Ω and not for $\mathbf{V}$ because Θ is defined on Ω^{op} (see Remark 2.14).

Remark 2.11. The map $\Theta(\psi) = \psi^* \times \text{id}$ is injective if and only if ψ is surjective, and it is surjective if and only if ψ is injective. Since μ_a appears in $P_{\mathbb{R},d}$ only for $a \in x(u)$, the mean μ is identified only on the observed levels, which is the analogue of the condition on the column rank under which β is identified in Example 2.12.

2.3.2 Multiple linear regression

Example 2.12.

1. **Units.** $\mathbf{U} = \text{FinSet}_{\text{inj}}$, as in Example 2.7.
2. **Covariate spaces.** $\Omega = \text{Vect}_{\mathbb{R}}^{\text{fd}}$. An object $\omega \cong \mathbb{R}^p$ is the space of the covariate vectors. A morphism $\psi: \omega \rightarrow \omega'$ is the inclusion of the covariates of a submodel into a larger set, or a linear recoding such as centring, orthogonalisation or contrast coding.
3. **Response scales.** $\mathbf{V}$ has the single object $\mathbb{R}$, with its Borel σ -algebra, and only the identity morphism (see Remark 2.14).
4. **Sample spaces.** $\mathbf{S} = \mathbf{V} \times \mathbf{U}^{\text{op}}$ and $\Gamma(\mathbb{R}, u) = \mathbb{R}^u$ with the product σ -algebra. The morphism $\varphi^{\text{op}}: u' \rightarrow u$ is mapped to the restriction $\text{res}_\varphi: \mathbb{R}^{u'} \rightarrow \mathbb{R}^u$.
5. **Designs.** An object (u, ω, x) of $\mathbf{D}$ is a design matrix, i.e. $x(i) \in \omega$ is the covariate vector of the unit i . Once a basis of $\omega \cong \mathbb{R}^p$ and an enumeration of u are chosen, the map x is the $n \times p$ matrix X .

6. **Parameters.** K and U_K are as in Example 2.7. The functor U_K is faithful because a linear map is determined by its underlying function. The parameter functor is the dual, i.e. the subrepresentation of the linear functionals inside the standard representation of McCullagh [21, §4.2, p. 1235],

$$\Theta(\omega) = \omega^* \times \mathbb{R}_{>0}, \quad \Theta(\psi)(\beta', \sigma^2) = (\beta' \circ \psi, \sigma^2).$$

For $d = (u, \omega, x)$ and $\theta = (\beta, \sigma^2) \in \omega^* \times \mathbb{R}_{>0}$ we set

$$P_{\mathbb{R},d}(\cdot \mid \beta, \sigma^2) := \bigotimes_{i \in u} \mathcal{N}(\beta(x(i)), \sigma^2) \quad \text{on } \mathbb{R}^u,$$

i.e. $Y \sim \mathcal{N}_n(X\beta, \sigma^2 I_n)$ in coordinates. This model has no intercept. One is obtained by taking as Ω the category of the finite dimensional real affine spaces and affine maps, and as $\Theta(\omega)$ the product of the space of the affine functionals on ω with $\mathbb{R}_{>0}$. This is what the regression examples of [21, §6] effectively use.

Lemma 2.13. *The family P of Example 2.12 is a statistical model.*

Proof. The proof is the same as that of Lemma 2.8. Let $f = (\varphi, \psi): d \rightarrow d'$ and let $\theta' = (\beta', \sigma^2) \in \Theta(\omega')$. Then $P_{\mathbb{R},d}(\cdot \mid \Theta(\psi)\theta') = \bigotimes_{i \in u} \mathcal{N}(\beta'(\psi(x(i))), \sigma^2)$, and the pushforward of $P_{\mathbb{R},d'}(\cdot \mid \theta')$ along res_φ is $\bigotimes_{i \in u} \mathcal{N}(\beta'(\varphi(x(i))), \sigma^2)$. The two measures coincide by (1). $\square$

Remark 2.14. Since L does not depend on V , a morphism $\alpha: v \rightarrow v'$ different from the identity imposes (3) at the same value of the parameter. Let $\alpha(y) = ay + b$. Then the left hand side of (3) is $\bigotimes_i \mathcal{N}(a\beta(x(i)) + b, a^2\sigma^2)$, which is equal to $P_{\mathbb{R},d}(\cdot \mid \beta, \sigma^2)$ only if $(a, b) = (1, 0)$. Hence the Gaussian linear model is not natural over $V = \text{Aff}(\mathbb{R})$, and for this reason we take V discrete here and in Example 2.7. The affine equivariance of the model is the statement that the model is a fixed point of an action on the parameter. To express it in the hexagon of Brøns one needs a parameter functor defined on $V^{\text{op}} \times \Omega^{\text{op}}$ and not on Ω^{op} alone. Indeed the parameter of McCullagh is a subrepresentation of v^ω and it does change with the response scale [21, p. 1235], so that the independence of Θ from V comes from the formulation of Brøns. Thus affine equivariance is a mild generalisation of Definition 2.1 and not an instance of it. The equivariance under the relabelling of the treatments is instead an instance of naturality, because it acts through Ω^{op} .

2.3.3 Binary classification

Example 2.15.

1. **Units.** $U = \text{FinSet}_{\text{inj}}$, as above.
2. **Covariate spaces.** $\Omega = \text{Vect}_{\mathbb{R}}^{\text{fd}}$. The objects are feature spaces, and the morphisms are linear feature maps.
3. **Response scales.** V has the two objects $\mathbf{2} = \{0, 1\}$ and $\mathbf{1} = \{*\}$, and its morphisms are the identities and the unique map $!: \mathbf{2} \rightarrow \mathbf{1}$. Thus the label can be discarded and cannot be relabelled. Indeed the involution $y \mapsto 1 - y$ carries $P_{\mathbf{2},d}(\cdot \mid \beta)$ to $P_{\mathbf{2},d}(\cdot \mid -\beta)$, hence it does not satisfy (3). It is an equivariance in the sense of Remark 2.14.
4. **Sample spaces.** $\Gamma(v, u) = v^u$ with the discrete σ -algebra, so that $\Gamma(\mathbf{2}, u) = \{0, 1\}^u$ is the space of the labels and $\Gamma(\mathbf{1}, u)$ is a point. The morphism φ^{op} is mapped to the restriction.
5. **Designs.** An object (u, ω, x) of D is the feature matrix. The labels are random and appear in Γ , not in D , i.e. the model is conditional on the features.
6. **Parameters.** $K = \text{Vect}_{\mathbb{R}}^{\text{fd}}$, U_K is the underlying-set functor, $\Theta(\omega) = \omega^*$ and $\Theta(\psi)(\beta') = \beta' \circ \psi$ for $\beta' \in \omega'^*$.

Let $\text{expit}(t) = (1 + e^{-t})^{-1}$. For $d = (u, \omega, x)$ and $\beta \in \Theta(\omega) = \omega^*$ we set

$$P_{2,d}(\cdot | \beta) := \bigotimes_{i \in u} \text{Bern}(\text{expit}(\beta(x(i)))) \quad \text{on } \{0, 1\}^u, \quad P_{1,d}(\cdot | \beta) := \delta \text{ on } \{*\}^u,$$

where δ is the unique probability measure on the point $\{*\}^u$.

Lemma 2.16. *The family P of Example 2.15 is a statistical model.*

Proof. For (2) at $v = 2$ the proof is the same as that of Lemma 2.8. Indeed, the marginal onto $\varphi(u)$ and the model at the pulled-back parameter are

$$\bigotimes_{i \in u} \text{Bern}(\text{expit}(\beta'(x'(\varphi(i))))) \quad \text{and} \quad \bigotimes_{i \in u} \text{Bern}(\text{expit}((\beta' \circ \psi)(x(i)))),$$

which coincide by (1). At $v = 1$ both sides are the unique probability measure on a point. For (3) at $! : 2 \rightarrow 1$, the pushforward of $P_{2,d}(\cdot | \beta)$ along $!^u : \{0, 1\}^u \rightarrow \{*\}^u$ is the Dirac measure, which is $P_{1,d}(\cdot | \beta)$ for every β . $\square$

2.3.4 Non-examples

The two families below are well-defined and indexed by the designs, but they are not statistical models. In both cases condition (N-U) fails.

Example 2.17. With the data of Example 2.12 we set

$$\tilde{P}_{\mathbb{R},d}(\cdot | \beta, \sigma^2) := \bigotimes_{i \in u} \mathcal{N}(\beta(x(i) - \bar{x}_d), \sigma^2), \quad \bar{x}_d = \frac{1}{|u|} \sum_{i \in u} x(i).$$

Then (2) fails already for $\psi = \text{id}$ and φ a proper inclusion. Indeed $\bar{x}_d \neq \bar{x}_{d'}$ in general, so that the mean of a given unit changes when other units are removed. The mean $\bar{x}_d$ is a statistic of the design, and the failure comes from its use in the definition of the family. The same mechanism underlies the non-example of Tjur (see Example 5.12).

Example 2.18. With the data of Example 2.15 we set

$$\tilde{P}_{2,d}(\cdot | \beta) := \bigotimes_{i \in u} \text{Bern}(s_i(\beta, d)), \quad s_i = \frac{e^{\beta(x(i))}}{\sum_{j \in u} e^{\beta(x(j))}},$$

i.e. a softmax taken across the sample and not across the label set. Then (2) fails as in Example 2.17, since removing a unit rescales all the remaining probabilities. Normalisation across the sample is a common operation in learning pipelines.

Remark 2.19. Summing up, condition (N-U) says that the marginal of the model on a subsample is the model of the subsample, and it excludes the normalisations which depend on the sample, as in Examples 2.17 and 2.18. Condition (N- Ω) says that the parameter is pulled back along the maps of covariate spaces, and it excludes the reparametrisations which depend on the design. Condition (N-V) expresses invariance at a fixed value of the parameter, and it excludes the relabelling and the rescaling of the response, which are equivariances (see Remark 2.14).

3 Markov-categorical background

We recall the few notions on Markov categories that we need. We refer the reader to [10, 23] for the general theory.

3.1 The Giry monad

Let $X = (X, \Sigma_X)$ be a measurable space. We denote by $\mathcal{G}X$ the set of the probability measures on X , endowed with the smallest σ -algebra such that the evaluation maps

$$\text{ev}_A: \mathcal{G}X \rightarrow [0, 1], \quad \text{ev}_A(\pi) = \pi(A),$$

are measurable for all $A \in \Sigma_X$. Let $f: X \rightarrow Y$ be a measurable map. We denote by $\mathcal{G}f: \mathcal{G}X \rightarrow \mathcal{G}Y$ the pushforward along f , i.e.

$$(\mathcal{G}f)(\pi)(B) = (f_*\pi)(B) = \pi(f^{-1}(B))$$

for all $\pi \in \mathcal{G}X$ and $B \in \Sigma_Y$. In this way we obtain a functor $\mathcal{G}: \text{Meas} \rightarrow \text{Meas}$. Let

$$\eta_X: X \rightarrow \mathcal{G}X, \quad \eta_X(x) = \delta_x,$$

where δ_x is the Dirac measure at x , and let

$$\mu_X: \mathcal{G}\mathcal{G}X \rightarrow \mathcal{G}X, \quad \mu_X(\Xi)(A) = \int_{\mathcal{G}X} \nu(A) \Xi(d\nu), \quad A \in \Sigma_X.$$

Thus μ_X maps a probability measure on the space of the probability measures on X to its mixture. Equivalently,

$$\int_X h(x) \mu_X(\Xi)(dx) = \int_{\mathcal{G}X} \left(\int_X h(x) \nu(dx) \right) \Xi(d\nu)$$

for every bounded measurable function $h: X \rightarrow \mathbb{R}$. The triple $(\mathcal{G}, \eta, \mu)$ is a monad on Meas , called the *Giry monad* (see [14]).

3.2 The category Stoch

A Kleisli morphism of the Giry monad from X to Y is a measurable map $k: X \rightarrow \mathcal{G}Y$. Equivalently, it is a *Markov kernel*, i.e. a map $k: X \times \Sigma_Y \rightarrow [0, 1]$ such that $B \mapsto k(x)(B)$ is a probability measure on Y for all $x \in X$, and $x \mapsto k(x)(B)$ is measurable for all $B \in \Sigma_Y$. The composition of $k: X \rightarrow \mathcal{G}Y$ and $k': Y \rightarrow \mathcal{G}Z$ is given by the Chapman–Kolmogorov formula

$$(k' \circ k)(x)(C) = \int_Y k'(y)(C) k(x)(dy), \quad C \in \Sigma_Z. \quad (4)$$

Definition 3.1. The Kleisli category of the Giry monad is denoted by Stoch . Its objects are the measurable spaces and its morphisms $f: X \rightsquigarrow Y$ are the Markov kernels, with the composition (4) (see [10, 14, 23]). We denote by FinStoch the full subcategory of Stoch of the finite discrete measurable spaces. Its morphisms are the stochastic matrices.

We write $\mathbf{A}(B, C)$ for the set of the morphisms from B to C in a category $\mathbf{A}$. By definition there is a bijection

$$\text{Stoch}(X, Y) \cong \text{Meas}(X, \mathcal{G}Y), \quad (5)$$

which is natural in X , via the precomposition with measurable maps, and in Y , via the pushforward.

The category Stoch is a *Markov category*, i.e. it is symmetric monoidal under the product of measurable spaces and it has copy and discard maps at every object, with naturality imposed on the discard maps alone [10, Def. 2.1]. We write $\text{copy}_X: X \rightarrow X \otimes X$ for the copy map at X .

Definition 3.2. A morphism $f: X \rightarrow Y$ of a Markov category is called *deterministic* if it is a homomorphism of the copy comonoids, i.e. if

$$\text{copy}_Y \circ f = (f \otimes f) \circ \text{copy}_X$$

(see [10, Def. 10.1]).

Thus f is deterministic if copying its output is the same as applying f independently to two copies of the input. In Stoch this means that $f(x)(B) \in \{0, 1\}$ for every $x \in X$ and every $B \in \Sigma_Y$, i.e. that each $f(x)$ is a $\{0, 1\}$ -valued measure [10, Ex. 10.4]. In FinStoch it means that the entries of the stochastic matrix of f belong to $\{0, 1\}$. The deterministic morphisms contain the identities and are closed under composition and under $\otimes$. Hence they form a wide symmetric monoidal subcategory $\text{Stoch}_{\text{det}} \subseteq \text{Stoch}$.

Definition 3.3. The *Dirac embedding* $\delta: \text{Meas} \rightarrow \text{Stoch}$ is the functor which is the identity on the objects and maps a measurable map f to the kernel $x \mapsto \delta_{f(x)}$, called the *Dirac kernel* of f . It is strong monoidal and its image is contained in $\text{Stoch}_{\text{det}}$, since a Dirac kernel is $\{0, 1\}$ -valued.

Lemma 3.4. Let Y be a measurable space.

1. The functor δ is injective on $\text{Meas}(X, Y)$ for all X if and only if Σ_Y separates points. This holds when Y is standard Borel, discrete or finite, and it does not hold for all Y .
2. If Σ_Y is countably generated and separates points, in particular if Y is standard Borel, then every deterministic morphism $X \rightsquigarrow Y$ is the Dirac kernel of a measurable map. This always holds in FinStoch .
3. Suppose that the codomains of the measurable maps f_j separate points. Then a diagram of Dirac kernels δf_j commutes in Stoch if and only if the diagram of the f_j commutes in Meas .

Proof. We have $\delta f = \delta g$ if and only if $\delta_{f(x)} = \delta_{g(x)}$ for all x , and $\delta_y = \delta_{y'}$ implies $y = y'$ for all y, y' if and only if Σ_Y separates points (for the converse take X a point). This gives (1), and (3) follows from (1) since δ is a functor.

For (2), let $(B_n)_n$ generate Σ_Y and let ν be a $\{0, 1\}$ -valued probability measure on Y . We set $C_n = B_n$ if $\nu(B_n) = 1$ and $C_n = Y \setminus B_n$ otherwise, and $C = \bigcap_n C_n$. Then $\nu(C) = 1$, hence $C \neq \emptyset$. Let $y, y' \in C$. The class of the sets $B \in \Sigma_Y$ which contain both or neither of y, y' is a σ -algebra which contains every B_n , hence it is Σ_Y , and $y = y'$ because Σ_Y separates points. Thus $C = \{y\}$ and $\nu = \delta_y$. Let now $k: X \rightsquigarrow Y$ be deterministic. Each $k(x)$ is $\{0, 1\}$ -valued [10, Ex. 10.4], hence $k(x) = \delta_{f(x)}$ for a unique $f(x) \in Y$. The map f is measurable because $f^{-1}(B) = \{x : k(x)(B) = 1\} \in \Sigma_X$ for every $B \in \Sigma_Y$, and $k = \delta f$. $\square$

Remark 3.5. The property of Lemma 3.4(2) says that the $\{0, 1\}$ -valued probability measures on Y are the points of Y . Conditions (a) and (b) of Appendix A are the two instances of this requirement which are used in Proposition A.4.

4 The bridge

In this section we prove Theorem 1.1. By (5) a component $P_{v,d}: U_K \Theta(\omega) \rightarrow \mathcal{P}(\Gamma(v, u))$ of a statistical model gives a morphism $\Theta(\omega) \rightsquigarrow \Gamma(v, u)$ of Stoch as soon as it is measurable in θ . This condition is empty in the finite case, which we consider first.

4.1 Finite case

Suppose that the design data of Definition 2.1 are *finite*, i.e. that the objects of $\mathbf{V}$, $\mathbf{\Omega}$ and $\mathbf{U}$ have finite underlying sets, that the spaces $\Gamma(v, u)$ are finite and discrete, and that $U_K \Theta(\omega)$ is finite for every ω . We endow each $U_K \Theta(\omega)$ with the discrete σ -algebra, so that $U_K \Theta$ lifts uniquely to a functor $\Theta_{\text{Meas}}: \mathbf{\Omega}^{\text{op}} \rightarrow \text{Meas}$ such that $U_K \Theta = |-| \circ \Theta_{\text{Meas}}$, where $|-|: \text{Meas} \rightarrow \text{Set}$ is the forgetful functor. We set

$$\mathbf{L} := \delta \circ \Theta_{\text{Meas}} \circ \text{pr}_{\mathbf{\Omega}}^{\text{op}} \circ \text{pr}_2, \quad \mathbf{R} := \delta \circ \Gamma \circ (\text{id}_{\mathbf{V}} \times \text{pr}_{\mathbf{U}}^{\text{op}}),$$

where δ is the Dirac embedding of Definition 3.3. These are two functors $\mathbf{V} \times \mathbf{D}^{\text{op}} \rightarrow \text{FinStoch}$.

Theorem 4.1. Suppose that the design data are finite. For a statistical model $P: L \Rightarrow R$ let $\mathbf{P}_{v,d}$ be the kernel $\theta \mapsto P_{v,d}(\cdot | \theta)$. Then $P \mapsto \mathbf{P}$ is a bijection between

1. the statistical models $P: L \Rightarrow R$ of Definition 2.2;
2. the natural transformations $\mathbf{P}: \mathbf{L} \Rightarrow \mathbf{R}$, i.e. the families of morphisms $\mathbf{P}_{v,d}: \mathbf{L}(v, d) \rightsquigarrow \mathbf{R}(v, d)$ of $\mathbf{FinStoch}$, indexed by the objects of $\mathbf{V} \times \mathbf{D}^{\text{op}}$, such that the square

$$\begin{array}{ccc}
 \mathbf{L}(v, d) & \rightsquigarrow^{\mathbf{P}_{v,d}} & \mathbf{R}(v, d) \\
 \mathbf{L}(m) \downarrow & & \downarrow \mathbf{R}(m) \\
 \mathbf{L}(v', d') & \rightsquigarrow^{\mathbf{P}_{v',d'}} & \mathbf{R}(v', d')
 \end{array} \tag{6}$$

commutes in $\mathbf{FinStoch}$ for every morphism $m: (v, d) \rightarrow (v', d')$ of $\mathbf{V} \times \mathbf{D}^{\text{op}}$.

The morphisms $\mathbf{L}(m)$ and $\mathbf{R}(m)$ are Dirac kernels, hence deterministic. The components $\mathbf{P}_{v,d}$ are arbitrary stochastic matrices, and $\mathbf{P}_{v,d}$ is deterministic if and only if $P_{v,d}(\cdot \mid \theta)$ is a Dirac measure for every θ .

Proof. Let $d = (u, \omega, x)$. Since δ is the identity on the objects we have $\mathbf{L}(v, d) = \Theta_{\text{Meas}}(\omega)$ and $\mathbf{R}(v, d) = \Gamma(v, u)$, which are finite and discrete. Recall that $U_K \Theta(\omega)$ is the underlying set of $\Theta_{\text{Meas}}(\omega)$ and that $\mathcal{P}(S)$ is the underlying set of $\mathcal{GS}$. Since $\Theta_{\text{Meas}}(\omega)$ is discrete, every map from it is measurable, hence

$$\text{Set}(U_K \Theta(\omega), \mathcal{P}(\Gamma(v, u))) = \text{Meas}(\Theta_{\text{Meas}}(\omega), \mathcal{G}\Gamma(v, u)).$$

Composing with (5) we obtain a bijection

$$\text{Set}(U_K \Theta(\omega), \mathcal{P}(\Gamma(v, u))) \xrightarrow{\cong} \mathbf{FinStoch}(\Theta_{\text{Meas}}(\omega), \Gamma(v, u)), \quad P_{v,d} \mapsto \mathbf{P}_{v,d}. \tag{7}$$

Let $k: X \rightsquigarrow Y$ be a kernel and let $g: Y \rightarrow Y'$ be a measurable map. By (4) we have

$$(\delta(g) \circ k)(x)(B) = \int \delta_{g(y)}(B) k(x)(dy) = k(x)(g^{-1}B) = (\mathcal{G}(g) \circ k)(x)(B), \tag{8}$$

i.e. the composition with $\delta(g)$ corresponds under (7) to the composition with the pushforward $\mathcal{P}(g) = \mathcal{G}(g)$. Let now $h: X' \rightarrow X$ be a measurable map. Then

$$(k \circ \delta(h))(x')(B) = \int k(x)(B) \delta_{h(x')}(dx) = k(h(x'))(B), \tag{9}$$

i.e. the precomposition with $\delta(h)$ is the precomposition with h .

Let $m = (\alpha, (\varphi, \psi)^{\text{op}})$ be a morphism of $\mathbf{V} \times \mathbf{D}^{\text{op}}$ with target (v', d') , where $d' = (u', \omega', x')$, so that $(\varphi, \psi): d' \rightarrow d$ is a morphism of $\mathbf{D}$ with $\varphi: u' \rightarrow u$ and $\psi: \omega' \rightarrow \omega$. We have $L(m) = U_K \Theta(\psi)$ and $R(m) = \mathcal{P}(\Gamma(\alpha, \varphi))$, while

$$\mathbf{L}(m) = \delta(\Theta_{\text{Meas}}(\psi)), \quad \mathbf{R}(m) = \delta(\Gamma(\alpha, \varphi)).$$

The naturality square of P at m is the equality

$$\mathcal{P}(\Gamma(\alpha, \varphi)) \circ P_{v,d} = P_{v',d'} \circ U_K \Theta(\psi)$$

of maps $U_K \Theta(\omega) \rightarrow \mathcal{P}(\Gamma(v', u'))$. By (8) the left hand side corresponds under (7) to $\mathbf{R}(m) \circ \mathbf{P}_{v,d}$, and by (9) the right hand side corresponds to $\mathbf{P}_{v',d'} \circ \mathbf{L}(m)$. Since (7) is a bijection, the naturality square of P at m commutes if and only if (6) commutes. Hence $P \mapsto \mathbf{P}$ is a bijection between the statistical models and the natural transformations $\mathbf{L} \Rightarrow \mathbf{R}$.

The morphisms $\mathbf{L}(m)$ and $\mathbf{R}(m)$ are Dirac kernels, hence they are deterministic by Definition 3.3. Finally $\mathbf{P}_{v,d}$ belongs to the image of δ if and only if $P_{v,d}(\cdot \mid \theta)$ is a Dirac measure for every θ , and the theorem is proved. $\square$

Remark 4.2. The discreteness of the parameter object is used only to obtain (7), while the finiteness of the spaces is used only to have $\mathbf{L}$, $\mathbf{R}$ and $\mathbf{P}$ in $\mathbf{FinStoch}$. The equality which gives (7) is the bijection of the adjunction $D \dashv U$ of Proposition A.5 at the pair $(U_K \Theta(\omega), \mathcal{G}\Gamma(v, u))$. By Theorem 4.1 all the notions available in $\mathbf{FinStoch}$ apply to the finite statistical models.

4.2 Measurable case

Definition 4.3. Suppose that the functor $U_K \Theta: \Omega^{\text{op}} \rightarrow \text{Set}$ lifts to a functor $\Theta_{\text{Meas}}: \Omega^{\text{op}} \rightarrow \text{Meas}$, i.e. that $|-| \circ \Theta_{\text{Meas}} = U_K \circ \Theta$. This means that each parameter object is endowed with a measurable structure and that the reparametrisation maps $\Theta(\psi)$ are measurable. A statistical model $P: L \Rightarrow R$ is called *measurable* if each component $P_{v,d}$ is measurable as a map $\Theta_{\text{Meas}}(\omega) \rightarrow \mathcal{G}\Gamma(v, u)$, i.e. if $\theta \mapsto P_{v,d}(A \mid \theta)$ is measurable for every measurable subset A of $\Gamma(v, u)$.

Let $\mathbf{L}, \mathbf{R}: \mathbf{V} \times \mathbf{D}^{\text{op}} \rightarrow \text{Stoch}$ be given as in the finite case, i.e. $\mathbf{L} = \delta \circ \Theta_{\text{Meas}} \circ \text{pr}_{\Omega}^{\text{op}} \circ \text{pr}_2$ and $\mathbf{R} = \delta \circ \Gamma \circ (\text{id}_{\mathbf{V}} \times \text{pr}_{\mathbf{U}}^{\text{op}})$.

Theorem 4.4. Let Θ_{Meas} be as in Definition 4.3. Then $P \mapsto \mathbf{P}$, where $\mathbf{P}_{v,d}$ is the kernel $\theta \mapsto P_{v,d}(\cdot \mid \theta)$, is a bijection between the measurable statistical models and the natural transformations $\mathbf{P}: \mathbf{L} \Rightarrow \mathbf{R}$ of functors $\mathbf{V} \times \mathbf{D}^{\text{op}} \rightarrow \text{Stoch}$, i.e. the families of morphisms $\mathbf{P}_{v,d}: \mathbf{L}(v, d) \rightsquigarrow \mathbf{R}(v, d)$ of Stoch such that the square (6) commutes in Stoch for every morphism m . The morphisms $\mathbf{L}(m)$ and $\mathbf{R}(m)$ are Dirac kernels, hence deterministic, for every morphism m . The components $\mathbf{P}_{v,d}$ need not be deterministic.

Proof. The bijection (5) and its naturality hold for arbitrary measurable spaces, and $P_{v,d}$ belongs to $\text{Meas}(\Theta_{\text{Meas}}(\omega), \mathcal{G}\Gamma(v, u))$ by hypothesis. The proof then goes as the one of Theorem 4.1, since (8) and (9) do not use finiteness. $\square$

Definition 4.5. The natural transformation $\mathbf{P}$ of Theorem 4.4 is called the *bridged model* associated to P .

Proof of Theorem 1.1. It follows from Theorem 4.1 and Theorem 4.4. $\square$

Remark 4.6. For a concrete model the hypothesis says that $\theta \mapsto P_{v,d}(A \mid \theta)$ is a measurable function for every event A . This is what is needed in order to integrate the model against a prior, and it holds when the model has a density which is jointly measurable in the observation and in the parameter. It is a restriction with respect to the formulation of McCullagh, which is Set -valued and imposes no measurability in θ . Hence Theorem 4.4 does not say that every statistical model in the sense of Definition 2.2 gives a natural transformation in Stoch . The hypothesis holds, with the obvious Borel structures, for every exponential family in its natural parametrisation and for all the examples of [21] with finite dimensional parameter spaces. The measure-valued and function-valued parameter spaces of [21, §8] require the choice of a σ -algebra, and we do not consider them.

Example 4.7. In Example 2.7 we have $\Theta(\omega) = \mathbb{R}^{\omega} \times \mathbb{R}_{>0}$ with its Borel structure, and the measurability of $(\mu, \sigma^2) \mapsto \bigotimes_i \mathcal{N}(\mu_{x(i)}, \sigma^2)$ is classical. Hence Theorem 4.4 applies, and the conditions of Remark 2.9 become commuting squares of kernels.

Remark 4.8 (Reduction to the finite case). The data of an experiment are finite, since responses and covariates are bounded by the range of the instrument and are recorded with finite precision. This does not reduce Theorem 4.4 to Theorem 4.1, but the measurable case can be approximated by finite ones in the following sense. Let $F \subset \mathbb{R}$ be a finite set, for example a set of floating-point numbers, and let $\rho: \mathbb{R} \rightarrow F$ be the rounding map, which is measurable and surjective. We add the object F and the morphism ρ to $\mathbf{V}$ and we set $P_{F,d}(\cdot \mid \theta) := (\rho^{\#})_* P_{\mathbb{R},d}(\cdot \mid \theta)$, so that (3) holds by construction. Hence the quantised model is the image of the model under a morphism of response scales, with no hypothesis and no error term. It is not an instance of Theorem 4.1. Indeed ρ does not change Θ , so that the parameter objects remain continua. A finite grid of parameter values is preserved by the maps $\Theta(\psi)$ when these are reindexings, as in the one-way layout. This is the setting of Section 6, and Proposition 6.11 gives the statement which the grids approximate. A grid is not preserved when $\Theta(\psi) = \psi^*$ uses the linear structure, as in Examples 2.12 and 2.15, because a floating-point grid is not closed under addition. Naturality is an equational condition, and we do not develop a notion of approximate naturality here. For this reason the computations of Section 6 are carried out in exact arithmetic. Note also that the pushforward of a Gaussian family along ρ is not an exponential family, and that the group means and the within-group sum of squares are not sufficient for it. Finally, let $(\rho_n: S \rightarrow F_n)_n$ be a refining sequence of finite quantisations whose cells

generate the σ -algebra of S . The canonical map $\mathcal{P}(S) \rightarrow \varprojlim_n \mathcal{P}(F_n)$ is injective, since the cells form a π -system, and it is not surjective in general. For example, for the dyadic quantisation of $[0, 1)$, the consistent family concentrated on the cells $[\frac{1}{2} - 2^{-n}, \frac{1}{2})$ has no preimage. Thus the measurable case is a projective limit of finite cases and not a special case of them.

4.3 The arrow category

In Appendix A we compare the category StatMod of Brøns with the category whose objects are the morphisms of Stoch and whose morphisms are the commuting squares with deterministic sides. The bridge is a computation with a representable functor and a copower; see (20). On the objects whose parameter set and sample space satisfy two regularity hypotheses, which always hold in the finite case, StatMod is equivalent to the category of the arrows of Stoch whose domain is a copower of the monoidal unit (Proposition A.4). Furthermore, StatMod is a coreflective subcategory of its measurable analogue (Proposition A.5). The sections that follow do not depend on the appendix.

5 Meaningfulness as naturality

In this section we prove Theorem 1.3. Recall that Tjur [26] considers a parameter function, i.e. a rule which assigns to each finite sample of covariate values a parameter of the corresponding model, in his words, a “function of the unknown distribution” [26, p. 1299]. He calls it *meaningful* when it is invariant under the formation of marginal models, i.e. when reducing the design by removing some of the sampled x ’s leaves the associated parameter unchanged. The criterion of McCullagh [21, §4.5] is that a subparameter be a natural transformation of functors on the design category. Thus Tjur quantifies over the removals of sample points and McCullagh over all the design morphisms.

5.1 Coherence

Let the design data be as in Definition 2.1. Recall that an *insertion* is a proper inclusion $\omega' \subsetneq \omega$ between objects of Ω (see Example 2.7). We denote by $\Omega_{\text{red}} \subseteq \Omega$ the subcategory generated by the insertions. Its morphisms are the inclusions, proper or not. We denote by $\Omega_{\text{inj}} \subseteq \Omega$ the subcategory generated by Ω_{red} and by the isomorphisms of Ω . In Example 2.7 its morphisms are the injections. An insertion leaves the surviving labels unchanged, while an injection which is not an inclusion is a relabelling followed by an insertion.

Definition 5.1. Let $F, G: \Omega^{\text{op}} \rightarrow \text{Stoch}$ be functors and let $\mathbf{Q} \subseteq \Omega$ be a subcategory which contains all the objects, called the *quantifier class*. A family $t = (t_\omega: F(\omega) \rightsquigarrow G(\omega))_{\omega \in \text{ob } \Omega}$ of morphisms of Stoch is called *$\mathbf{Q}$ -coherent* if

$$t_\omega \circ F(\psi) = G(\psi) \circ t_{\omega'} \quad \text{for every } \psi: \omega \rightarrow \omega' \text{ in } \mathbf{Q}, \quad (10)$$

i.e. if t is a natural transformation $F|_{\mathbf{Q}^{\text{op}}} \Rightarrow G|_{\mathbf{Q}^{\text{op}}}$.

Coherence over a class implies coherence over every subclass, and Ω -coherence is naturality. In words, a family is coherent over $\mathbf{Q}$ if every change of covariate space which belongs to $\mathbf{Q}$ carries the family to itself.

McCullagh takes the morphisms of Ω to be injective [21, §4.1]. Hence under his convention a merging is not a morphism, and the difference between his criterion and the one of Tjur is given by the bijections, i.e. by the relabellings. In Definition 2.1 we follow Brøns and we impose no such restriction. Since every map of covariate spaces is a surjection followed by an injection, in our setting there are three quantifier classes

$$\underbrace{\Omega_{\text{red}}}_{\text{Tjur}} \subseteq \underbrace{\Omega_{\text{inj}}}_{\text{McCullagh as stated}} \subseteq \underbrace{\Omega}_{\text{naturality here}},$$

which are strictly nested in Example 2.7. The first inclusion adds the relabellings and the second one adds the mergings. The corresponding conditions are ordered in the opposite way, i.e. coherence over Ω is the strongest one. Neither source asserts that the criterion of Tjur and the one of McCullagh coincide, and in general they do not. By Theorem 1.1 the three conditions are equalities between morphisms of the same category, so that they can be compared. In the one-way layout, the failure of the marginal dispersion is detected by the insertions, i.e. by the weaker criterion of Tjur (see Example 6.10).

5.2 Design-indexed quantities

Definition 5.2. Let $\mathbf{P}: \mathbf{L} \Rightarrow \mathbf{R}$ be a bridged model (Definition 4.5) and let $\Lambda: \Omega^{\text{op}} \rightarrow \mathbf{Meas}$ be a functor, called the *target*. The measurable space $\Lambda(\omega)$ is the space of values at ω . A *design-indexed quantity* for $\mathbf{P}$ with target Λ is a family of morphisms of $\mathbf{Stoch}$

$$g_\omega: \Theta_{\text{Meas}}(\omega) \rightsquigarrow \Lambda(\omega), \quad \omega \in \text{ob } \Omega.$$

It is called *deterministic* if $g_\omega = \delta f_\omega$ for measurable maps $f_\omega: \Theta_{\text{Meas}}(\omega) \rightarrow \Lambda(\omega)$, i.e. if it is an ordinary functional of the parameter. The quantity g is called *meaningful* if it is Ω -coherent in the sense of Definition 5.1 with $F = \delta \Theta_{\text{Meas}}$ and $G = \delta \Lambda$, i.e. if

$$g_\omega \circ \delta \Theta_{\text{Meas}}(\psi) = \delta \Lambda(\psi) \circ g_{\omega'} \quad \text{in } \mathbf{Stoch} \quad (11)$$

for every $\psi: \omega \rightarrow \omega'$ in Ω . It is called *Tjur-natural* if it is Ω_{red} -coherent, i.e. if (11) holds for every ψ in Ω_{red} .

Thus a meaningful quantity is a natural transformation $\delta \Theta_{\text{Meas}} \Rightarrow \delta \Lambda$ of functors $\Omega^{\text{op}} \rightarrow \mathbf{Stoch}$. The functors are the same for the deterministic quantities and for the general ones, and only the components differ.

Example 5.3. In the one-way layout (Example 2.7) we have $\Theta(\omega) = \mathbb{R}^\omega \times \mathbb{R}_{>0}$ and $\Theta(\psi)(\mu', \sigma^2) = (\mu' \circ \psi, \sigma^2)$. The vector of the group means $g_\omega(\mu, \sigma^2) = \mu$ has target $\Lambda(\omega) = \mathbb{R}^\omega$ with $\Lambda(\psi) = \psi^*$, i.e. the restriction $\mu' \mapsto \mu' \circ \psi$. The variance $g_\omega(\mu, \sigma^2) = \sigma^2$ has the constant target $\Lambda(\omega) = \mathbb{R}_{>0}$ with $\Lambda(\psi) = \text{id}$. Both are meaningful. Indeed (11) reads $\mu' \circ \psi = \psi^* \mu'$ in the first case and $\sigma^2 = \sigma^2$ in the second one.

Remark 5.4. The components of McCullagh in [21, §4.5] are indexed by the designs and are natural over all the design morphisms (ψ_d, ψ_c) , while in Definition 5.2 the quantities are indexed by the objects of Ω . The two indexings agree on the families which factor through pr_Ω . Indeed, for these families the squares with identity covariate leg are trivial. Furthermore, for every $\psi: \omega \rightarrow \omega'$ and every design (u, ω, x) the pair $(\text{id}_u, \psi): (u, \omega, x) \rightarrow (u, \omega', \psi \circ x)$ is a morphism of $\mathbf{D}$, so that pr_Ω is surjective on the morphisms.

5.3 Tjur's criterion

In this subsection $\mathbf{C} = \mathbf{Set}$, the functors J_U and J_Ω are the underlying-set functors; Ω is a category of finite sets such as Ω_{lab} , and $\mathbf{U}$ is a full subcategory of $\mathbf{FinSet}_{\text{inj}}$ which contains the underlying sets of the objects of Ω , so that every inclusion of a subsample between such sets is a morphism of $\mathbf{U}$. The regression examples of Section 2.3, whose covariate spaces are vector spaces, do not satisfy these hypotheses.

Definition 5.5. Let (u, ω, x) be a design with x surjective. A *Tjur reduction* of (u, ω, x) is a morphism of $\mathbf{D}$

$$(\varphi, \psi): (u', \omega', x') \longrightarrow (u, \omega, x)$$

where $\varphi: u' \hookrightarrow u$ is the inclusion of a proper subsample, $\omega' = x(\varphi(u'))$ is the surviving support, x' is the corestriction of $x \circ \varphi$, and $\psi: \omega' \hookrightarrow \omega$ is the inclusion.

A Tjur reduction is the removal of the sample points $u \setminus \varphi(u')$ together with the covariate values which are no longer observed, i.e. Tjur's “removal of some of the x 's” [26, p. 1299]. His condition at such a reduction compares the parameter assigned to the marginal model with the parameter of the bigger

sample, the former being for him a function of the marginal distribution. Since a bridged model is natural at (φ, ψ) , see (2), the marginal law at the reduced design depends on θ only through $\Theta_{\text{Meas}}(\psi)\theta$. The values at ω are carried to ω' by $\Lambda(\psi)$, which is the identity in Tjur's setting, where the scale is fixed. Hence the comparison is the square (11) at ψ , and we say that g satisfies Tjur's condition at the reduction if that square commutes.

Lemma 5.6. *A design-indexed quantity g satisfies Tjur's condition at every Tjur reduction of every design with surjective design map if and only if g is Tjur-natural.*

Proof. Suppose that g is Tjur-natural. The covariate leg ψ of a Tjur reduction is the inclusion of the surviving support. It is an insertion when the removal empties some fibre of x , and it is the identity otherwise. The square (11) holds in the first case by hypothesis and in the second one trivially.

Conversely, it is enough to check (11) at each insertion $\psi: \omega' \subseteq \omega$, since such squares can be pasted along the compositions and hold trivially at the identities. Consider the tautological design $(\omega, \omega, \text{id}_\omega)$, which has one unit for each label and surjective design map, and the subsample $\varphi = \psi: \omega' \hookrightarrow \omega$. It is a morphism of $\mathbf{U}$ because ω' and ω are objects of Ω , hence of $\mathbf{U}$, and $\mathbf{U}$ is full in $\text{FinSet}_{\text{inj}}$. The surviving support is $x(\varphi(\omega')) = \omega'$, so that $(\psi, \psi): (\omega', \omega', \text{id}) \rightarrow (\omega, \omega, \text{id})$ is a Tjur reduction with covariate leg ψ , and Tjur's condition at it is the square (11) at ψ . $\square$

The parameter functions of Tjur are indexed by the finite samples of covariate values, with multiplicities, and not by the covariate spaces, while Lemma 5.6 concerns quantities which are already indexed by the support. We now remove this restriction.

Definition 5.7. A sample-indexed quantity for $\mathbf{P}$ with target $\Lambda: \Omega^{\text{op}} \rightarrow \text{Meas}$ is a family of morphisms of Stoch

$$G_d: \Theta_{\text{Meas}}(\omega) \rightsquigarrow \Lambda(\omega), \quad d = (u, \omega, x) \text{ a design with } x \text{ surjective,}$$

i.e. one morphism for each finite sample of covariate values, where the sample is the pair (u, x) and ω is its support. It is called *Tjur-meaningful* if

$$G_{d'} \circ \delta\Theta_{\text{Meas}}(\psi) = \delta\Lambda(\psi) \circ G_d \quad \text{in } \text{Stoch} \quad (12)$$

for every Tjur reduction $(\varphi, \psi): d' \rightarrow d$. It is called *support-indexed* if $G_{(u, \omega, x)} = g_\omega$ for a design-indexed quantity g , i.e. if it depends on the sample only through its support.

Example 5.8. Let Λ be constant at $\mathbb{R}$. The sample size $G_{(u, \omega, x)} = |u|$ is a sample-indexed quantity. In the simple regression model of Example 5.12, the covariate average weighted by the multiplicities gives the sample-indexed quantity

$$G_{(u, \omega, x)}(\alpha, \beta, \sigma^2) = \alpha + \beta |u|^{-1} \sum_{i \in u} x(i).$$

These are the examples of Tjur [26, pp. 1299–1300], and neither of them is support-indexed.

Proposition 5.9. *Suppose that $\mathbf{U}$ contains a disjoint union $u \sqcup u'$ of any two of its objects, as $\text{FinSet}_{\text{inj}}$ does. For a sample-indexed quantity G with target Λ , the following are equivalent:*

1. G is Tjur-meaningful;
2. G is support-indexed, i.e. $G_{(u, \omega, x)} = g_\omega$, and the design-indexed quantity g is Tjur-natural.

The quantity g of part (2) is unique, namely $g_\omega = G_{(\omega, \omega, \text{id}_\omega)}$. In particular a Tjur-meaningful sample-indexed quantity depends neither on the sample size nor on the multiplicities.

Proof. (2) $\Rightarrow$ (1). Let $(\varphi, \psi): d' \rightarrow d$ be a Tjur reduction. The covariate leg ψ is the inclusion of the surviving support, i.e. an insertion or an identity, and (12) is the square (11) at ψ . It holds in the first case because g is Tjur-natural and in the second one trivially.

(1) $\Rightarrow$ (2). Let $d = (u, \omega, x)$ and $d' = (u', \omega, x')$ be two designs with the same support $\omega \neq \emptyset$, so that u and u' are non-empty. We set $d'' = (u \sqcup u', \omega, [x, x'])$, whose design map is surjective because x is. The

two injections of the coproduct are inclusions of proper subsamples into $u \sqcup u'$, the surviving support of each of them is ω , and the corestrictions are x and x' . Hence $(\iota_u, \text{id}_\omega): d \rightarrow d''$ and $(\iota_{u'}, \text{id}_\omega): d' \rightarrow d''$ are Tjur reductions with identity covariate leg. At such a reduction (12) reads $G_d = G_{d''}$, since Θ_{Meas} and Λ map identities to identities. Therefore $G_d = G_{d''} = G_{d'}$, i.e. G depends on the sample only through its support. For $\omega = \emptyset$ there is nothing to prove. We set $g_\omega = G_{(\omega, \omega, \text{id}_\omega)}$, where the tautological design exists because ω is an object of $\mathbf{U}$. Let $\psi: \omega' \subseteq \omega$ be an insertion. The tautological reduction $(\psi, \psi): (\omega', \omega', \text{id}) \rightarrow (\omega, \omega, \text{id})$ of the proof of Lemma 5.6 is a Tjur reduction with covariate leg ψ , and (12) at it is the square (11) for g at ψ . Hence g is Tjur-natural. The uniqueness follows from the formula for g_ω , and the last statement follows from part (2). $\square$

Thus Tjur's criterion, imposed on the quantities that he considers, first forces the indexing by the support and then reduces to Tjur-naturality in the sense of Definition 5.2.

Remark 5.10. The parameter functions of Tjur are functions of the unknown distribution [26, p. 1299], while our quantities are functions of the parameter. Every function of the distribution is a function of the parameter, and the converse holds when $\theta \mapsto \mathbf{P}_{\mathbf{v},d}(\cdot \mid \theta)$ is injective. We do not pursue this distinction.

5.4 Meaningful and Tjur-natural quantities

Proposition 5.11. *Let g be a design-indexed quantity.*

1. *Suppose that g is deterministic, with $g_\omega = \delta f_\omega$, and that the σ -algebras of the spaces $\Lambda(\omega)$ separate points. Then g is meaningful if and only if the maps f_ω form a natural transformation $\Theta_{\text{Meas}} \Rightarrow \Lambda$ in Meas .*
2. *If g is meaningful then it is Tjur-natural. The converse holds when $\Omega_{\text{red}} = \Omega$ and it does not hold in general (see Example 5.13).*
3. *Meaningfulness is preserved by the composition with the natural transformations of targets $\Lambda \Rightarrow \Lambda'$ (in Meas , or in Stoch for the general quantities). It is also preserved by the pullback along the functors $F: \Omega' \rightarrow \Omega$ such that $J_{\Omega'} = J_\Omega \circ F$, i.e. by a change of the covariate scheme. In this case $(u, \omega', x) \mapsto (u, F\omega', x)$ is a functor $\mathbf{D}' \rightarrow \mathbf{D}$, along which the bridged model $\mathbf{P}$ is restricted.*

Proof. (1) Let $g = \delta f$. Then (11) reads $\delta(f_\omega \circ \Theta_{\text{Meas}}(\psi)) = \delta(\Lambda(\psi) \circ f_{\omega'})$. Since δ is injective on the maps into $\Lambda(\omega)$ by Lemma 3.4(1), this holds if and only if $f_\omega \circ \Theta_{\text{Meas}}(\psi) = \Lambda(\psi) \circ f_{\omega'}$.

(2) Tjur-naturality is (11) for the morphisms of Ω_{red} only. This gives the first two statements. The quantity of Example 5.13 is Tjur-natural and not meaningful. Note that the quantity of Example 5.12 fails already on Ω_{red} , so that it does not distinguish the two conditions.

(3) The first statement follows by pasting the squares. For the pullback, $g_{F\omega'}$ is natural in ω' because F maps the morphisms of Ω' to morphisms of Ω . The hypothesis $J_{\Omega'} = J_\Omega F$ gives that the restricted model is a bridged model over Ω' , since a design (u, ω', x) for Ω' has $x: J_\Omega u \rightarrow J_{\Omega'} \omega' = J_\Omega F\omega'$. $\square$

Example 5.12. We write the non-example of Tjur [26, pp. 1299–1300] in our notation. Consider the simple regression model with $\Theta(\omega) = \mathbb{R}^2 \times \mathbb{R}_{>0}$ and coordinates $(\alpha, \beta, \sigma^2)$, let Ω be the category of the finite selections of covariate values, and let Λ be constant at $\mathbb{R}$ with $\Lambda(\psi) = \text{id}$. This is the unreplicated case, where the design map is injective, so that sample points and covariate values coincide. The general case, where $\bar{x}$ is weighted by the multiplicities, is the sample-indexed quantity of Example 5.8, which is excluded by Proposition 5.9 before any comparison between support-indexed quantities is made. We set

$$g_\omega(\alpha, \beta, \sigma^2) = \alpha + \beta \bar{x}_\omega, \quad \bar{x}_\omega = |\omega|^{-1} \sum_{x \in \omega} x.$$

Each g_ω is a parameter of the model at ω . Let $\psi: \omega \subseteq \omega'$ be an insertion. The parameter map $\Theta(\psi)$ is the identity, while $\bar{x}_\omega \neq \bar{x}_{\omega'}$ in general. Hence (11) reads $\alpha + \beta \bar{x}_\omega = \alpha + \beta \bar{x}_{\omega'}$, and it fails when $\beta \neq 0$. Since the failure occurs on Ω_{red} , the quantity is neither Tjur-natural nor meaningful. The same argument, where $\Theta(\psi)$ is again the identity, applies to the correlation coefficient, which depends on the design through the variance of the covariate [21, §6.2]. In this case it is the scale of the design, and not its mean, which is not preserved. In the bridged setting, both failures are equalities which do not hold.

Example 5.13. Let $\Omega = \Omega_{\text{lab}}$ (Example 2.7), let Θ be any parameter functor, and let $\Lambda(\omega) = \mathbb{R}^{\omega \times \omega}$ with $\Lambda(\psi) = (\psi \times \psi)^*$, i.e. $\Lambda(\psi)(v)_{(a,b)} = v_{(\psi a, \psi b)}$. We set $g_\omega(\theta)_{(a,b)} = [a \neq b]$, the indicator that two treatment labels are distinct. The square (11) at $\psi: \omega \rightarrow \omega'$ reads $[a \neq b] = [\psi a \neq \psi b]$ for all $a, b \in \omega$, which holds if and only if ψ is injective. Hence g is natural along every injection. In particular it is Tjur-natural, and it satisfies the criterion of McCullagh as stated. On the other hand, g is not natural along any map which is not injective, for example along $p: \{1, 2\} \rightarrow \{*\}$. Thus the difference between Tjur-naturality and meaningfulness occurs at the mergings.

Proof of Theorem 1.3. It follows from Propositions 5.9 and 5.11(2) and from Example 5.13. $\square$

Corollary 5.14. *Suppose that Ω has finitely many morphisms, that $\Theta_{\text{Meas}}(\omega)$ and $\Lambda(\omega)$ are finite and discrete for every ω , and that the entries of the kernels g_ω belong to a subfield $k \subseteq \mathbb{R}$ with decidable equality, for example $\mathbb{Q}$ or the field of the real algebraic numbers. Then the meaningfulness of g is decidable. Indeed, it is equivalent to the finite conjunction, over the morphisms $\psi: \omega \rightarrow \omega'$ of any generating set of Ω , of the matrix equalities (11), each of which is an equality between two stochastic matrices of size $|\Theta_{\text{Meas}}(\omega')| \times |\Lambda(\omega)|$ with entries in k .*

Proof. If (11) holds at ψ and at ψ' then it holds at $\psi'\psi$, by pasting the two squares, since $\delta\Theta_{\text{Meas}}$ and $\delta\Lambda$ are functors. Hence it is enough to check a generating set, which is finite, and each check is an equality between finitely many elements of k , which is decidable by hypothesis. $\square$

6 A cohomological formulation of coherence for finite designs

In this section we prove Theorem 1.5. We suppose that the design scheme is finite, and we show that the failure of naturality of a quantity is a 1-cochain of a complex of Baues and Wirsching. This links the present framework to the diagrammatic Hochschild cohomology of Gerstenhaber and Schack and to the cohomology of small categories of Baues and Wirsching [12, 2], as revisited in [6, §3]. The cohomology controls the obstructions to the Ω -coherence of the quantities (Definition 5.1).

6.1 The diagram of algebras of a finite design scheme

Let Ω be a finite category of covariate spaces and suppose that the parameter sets $\Theta(\omega)$ and the target sets $\Lambda(\omega)$ are finite, as in Corollary 5.14. Let $k \supseteq \mathbb{Q}$ be a field.

Definition 6.1. We denote by Alg_k the category of associative unital k -algebras and unital homomorphisms. The *parameter diagram of algebras* is the functor

$$A: \Omega \longrightarrow \text{Alg}_k, \quad A(\omega) := k^{\Theta(\omega)}, \quad A(\psi) := \Theta(\psi)^*,$$

where $k^{\Theta(\omega)}$ is the algebra of the functions on the parameter set, with the pointwise operations, and $\Theta(\psi)^*$ is the precomposition with $\Theta(\psi)$. It is covariant because Θ and $k^{(-)}$ are both contravariant. In the same way a target Λ gives the functor $B(\omega) := k^{\Lambda(\omega)}$.

Let $g = (g_\omega: \Theta(\omega) \rightarrow \Lambda(\omega))$ be a deterministic design-indexed quantity. It gives a family of homomorphisms of algebras $g^* = (g_\omega^*: B(\omega) \rightarrow A(\omega))$, and g is meaningful (Definition 5.2) if and only if g^* is a natural transformation $B \Rightarrow A$. Indeed, all the spaces are finite and discrete, so that meaningfulness is the naturality of the underlying maps by Proposition 5.11(1). Furthermore $k^{(-)}$ is faithful, since a map f is recovered from f^* through the indicator functions of its fibres. Hence g is natural if and only if g^* is. For $\psi: \omega \rightarrow \omega'$ we set

$$c(\psi) := A(\psi)g_\omega^* - g_{\omega'}^*B(\psi) \in \text{Hom}_k(B(\omega), A(\omega')), \quad (13)$$

and we call $c = c_g$ the *defect* of g . It measures the failure of naturality of g at ψ .

The defect is a 1-cochain on Ω in the sense of Baues and Wirsching [2, (1.4)], with coefficients in a natural system $D: F\Omega \rightarrow \text{Ab}$ in their sense [2, (1.2)]. Here $F\Omega$ is the category of factorisations of Ω , i.e.

the twisted arrow category, denoted by $\text{Tw } \Omega$ in [6]. Its objects are the morphisms of Ω , and a morphism $\psi \rightarrow \psi'$ is a pair (α, β) such that $\psi' = \alpha\psi\beta$. The natural system that we use is obtained by pulling back the bimodule $M(\omega, \omega') = \text{Hom}_k(B(\omega), A(\omega'))$ along the forgetful functor $F\Omega \rightarrow \Omega^{\text{op}} \times \Omega$ [2, (1.16)–(1.18)]. Thus $D(\psi) = \text{Hom}_k(B(\omega), A(\omega'))$ for $\psi: \omega \rightarrow \omega'$, and $(\alpha, \beta): \psi \rightarrow \alpha\psi\beta$ acts by $f \mapsto A(\alpha)fB(\beta)$. With bimodule coefficients, this is the Hochschild–Mitchell cohomology [2, (8.5)]. We write

$$C^0 = \prod_{\omega} \text{Hom}_k(B(\omega), A(\omega)), \quad (d^0 h)(\psi) = A(\psi)h_{\omega} - h_{\omega'}B(\psi),$$

for the group of the 0-cochains and for the differential. The equations $d^0 h = 0$, one for each ψ , define the end [20, IX.5]

$$\text{Nat}(B, A) = \int_{\omega} \text{Hom}_k(B(\omega), A(\omega)) = \ker d^0 = H^0, \quad (14)$$

and the defect (13) of a quantity g is the coboundary $c_g = d^0 g^*$. By [2, (5.3)–(5.6)] the group H^1 is the group of the derivations of Ω into D , i.e. of the 1-cochains c such that $c(\psi'\psi) = A(\psi')c(\psi) + c(\psi')B(\psi)$, modulo the inner derivations $d^0 h$. We write Z^1 and B^1 for the groups of the 1-cocycles and of the 1-coboundaries.

Remark 6.2. When $B = A$, the complex above is the row of Hochschild degree 1 of the Gerstenhaber–Schack bicomplex of the diagram A [6, Prop. 3.8]. This bicomplex is defined over a poset in [12], for diagrams over a small category as in [13, §2], and over a general finite category in [6, eq. (4)]. Here A is read as a presheaf on Ω^{op} , which is the convention of [6], and the Baues–Wirsching complexes of Ω and of Ω^{op} with the transported natural system are isomorphic up to sign, since $F(\Omega^{\text{op}}) \cong F\Omega$. The group H^1 computed below is the cohomology of that row for its horizontal differential, which is the Baues–Wirsching one, i.e. it lives in bidegree $(1, 1)$ of the bicomplex. It is not the Gerstenhaber–Schack cohomology in total degree 2, and no vertical (Hochschild) differential is used here. No result of [6] enters a proof in this paper. That work is cited for the identification just recalled, which concerns the case $B = A$ and not the case computed below, and for the convention on the twisted arrow category. The relation of the bicomplex to the Baues–Wirsching cohomology is controlled by a spectral sequence attributed to Robinson [25] in [6, §1, Prop. 3.10 and Thm. 3.11]. That sequence takes the vertical differential first, so that its E_2 page has Hochschild cohomology groups as coefficients, and we do not use it here.

Lemma 6.3. *The cochain c of (13) satisfies the 1-cocycle identity*

$$c(\psi'\psi) = A(\psi')c(\psi) + c(\psi')B(\psi)$$

for all composable ψ, ψ' , and $c = 0$ if and only if g is meaningful.

Proof. We have $c = d^0 g^*$ by (13), and the identity is $d^1 d^0 = 0$. Explicitly, since A and B are functors, adding and subtracting the term $A(\psi')g_{\omega'}^* B(\psi)$ we obtain

$$A(\psi'\psi)g_{\omega}^* - g_{\omega'}^* B(\psi'\psi) = A(\psi')(A(\psi)g_{\omega}^* - g_{\omega'}^* B(\psi)) + (A(\psi')g_{\omega'}^* - g_{\omega'}^* B(\psi'))B(\psi).$$

Finally $c = 0$ means that g^* is natural, which holds if and only if g is natural, as we have seen after Definition 6.1. $\square$

Let g be a deterministic design-indexed quantity, given on the objects of Ω only. We call it a *raw family*. An *admissible class* is a subset $C \subseteq C^0$, and a *correction* of g in C is an element $h \in C$ such that $g^* + h$ is natural.

Proposition 6.4. *Let g be a raw family and let $C \subseteq C^0$ be an admissible class.*

1. *There is a correction of g in C if and only if $c_g = -d^0 h$ for some $h \in C$.*
2. *We have $c_g \in B^1$ for every raw family, hence $[c_g] = 0$ in H^1 with no hypothesis. If $C = C^0$ then $h = -g^*$ is always a correction.*

3. Suppose that C is a linear subspace. The obstruction to the existence of a correction of g in C is the class of g^* in $C^0/(C + \text{Nat}(B, A))$. The differential d^0 identifies this quotient with $B^1/d^0(C)$, i.e. with the cokernel of d^0 restricted to C and corestricted to B^1 . The image of the obstruction in $H^1 = Z^1/B^1$ is zero.

Proof. Let $h \in C^0$. Then $g^* + h$ is natural if and only if $d^0(g^* + h) = 0$, i.e. if and only if $c_g = -d^0h$. This gives part (1). Part (2) follows from $c_g = d^0g^*$ and from the case $h = -g^*$. For part (3), we have $c_g = -d^0h$ with $h \in C$ if and only if $g^* \in C + \ker d^0 = C + \text{Nat}(B, A)$. The map d^0 induces an isomorphism $C^0/\ker d^0 \cong B^1$ which carries $C + \ker d^0$ onto $d^0(C)$, and B^1 is the denominator of H^1 . $\square$

Remark 6.5. A 0-cochain is a family of linear maps. Hence $g^* + h$ is the dual of a design-indexed quantity only when it is a family of homomorphisms of algebras, and this has to be ensured by C . The class of the recalibrations of Example 6.10 does so by construction. By part (2) an admissible class must be a proper subset of C^0 in order to carry an obstruction. By part (3) the group $H^1(\Omega; \text{Hom}_k(B, A))$ is not an obstruction group for the quantities. It is an invariant of the design scheme and of the target Λ , and Example 6.12 gives a case where it is not zero. All the computations are finite because the complex is finite dimensional in each degree. For a linear subspace C , they are linear algebra, and for a finite class which is not linear, such as the recalibrations, they are combinatorics (Proposition 6.9). Both c_g and the correction h , when it exists, are computed exactly.

The obstruction of part (3) becomes a cohomology class when the complex is taken relative to the admissible class.

Proposition 6.6. Let $C^\bullet = C^\bullet(\Omega; D)$ be the Baues–Wirsching complex introduced above and let $C \subseteq C^0$ be a linear subspace. Let $C_C^\bullet \subseteq C^\bullet$ be the subcomplex with $C_C^0 = C$ and $C_C^n = C^n$ for $n \geq 1$. Then

$$H^0(C_C^\bullet) = C \cap \text{Nat}(B, A), \quad H^1(C_C^\bullet) = Z^1/d^0(C), \quad H^n(C_C^\bullet) = H^n(\Omega; \text{Hom}_k(B, A)) \text{ for } n \geq 2,$$

and the short exact sequence of complexes $0 \rightarrow C_C^\bullet \rightarrow C^\bullet \rightarrow C^0/C \rightarrow 0$, where the quotient is concentrated in degree 0, gives the exact sequence

$$0 \rightarrow C \cap \text{Nat}(B, A) \rightarrow \text{Nat}(B, A) \rightarrow C^0/C \xrightarrow{\delta} H^1(C_C^\bullet) \rightarrow H^1(\Omega; \text{Hom}_k(B, A)) \rightarrow 0 \quad (15)$$

with connecting map $\delta[h] = [d^0h]$. Furthermore:

1. for a raw family g we have $\delta[g^*] = [c_g]_C$, the class of the defect in $H^1(C_C^\bullet)$, and g has a correction in C if and only if $[c_g]_C = 0$. The correction is unique modulo $C \cap \text{Nat}(B, A)$, hence it is unique when $C \cap \text{Nat}(B, A) = 0$;
2. the group $B^1/d^0(C)$ of Proposition 6.4(3) is the image of δ , which is the kernel of the comparison map $H^1(C_C^\bullet) \rightarrow H^1(\Omega; \text{Hom}_k(B, A))$. The absolute group is the cokernel of δ , i.e. the part of $H^1(C_C^\bullet)$ which is not reached by any quantity.

Proof. Note that $C_C^\bullet$ is a subcomplex of $C^\bullet$ because $d^0(C) \subseteq C^1$ and its terms of positive degree are those of $C^\bullet$. The quotient is C^0/C in degree 0 and zero elsewhere, with zero differential. The cohomology of the subcomplex is $C \cap \ker d^0$ in degree 0, $Z^1/d^0(C)$ in degree 1 and $H^n(C^\bullet)$ for $n \geq 2$, and $\ker d^0 = \text{Nat}(B, A)$ by (14). The long exact sequence of the short exact sequence of complexes is (15), since the cohomology of the quotient is zero in positive degrees. Its connecting map lifts $[h] \in C^0/C$ to $h \in C^0$ and applies d^0 , whose values belong to $C^1 = C_C^1$, hence $\delta[h] = [d^0h]$. The exactness at C^0/C says that $\delta[h] = 0$ if and only if $h \in C + \text{Nat}(B, A)$. The exactness at $H^1(C_C^\bullet)$ says that the image of δ , which is $B^1/d^0(C)$, is the kernel of the comparison map, which is surjective because Z^1 maps onto Z^1/B^1 . This gives part (2). Part (1) follows from the exactness at C^0/C applied to $h = g^*$, since $d^0g^* = c_g$, and from Proposition 6.4(3). If $h, h' \in C$ are two corrections of the same g then $g^* + h$ and $g^* + h'$ are natural, hence $h - h' \in C \cap \text{Nat}(B, A)$. $\square$

6.2 The one-way layout as test case

We now compute the objects introduced above for the one-way layout. We first give the setting (Example 6.7), then we describe what is computed and how, and finally we give the results (Propositions 6.8, 6.9 and 6.11, Examples 6.10 and 6.12).

Example 6.7. We restrict Example 2.7 to two covariate spaces, namely $\omega_2 = \{1, 2\}$ (two treatment groups) and $\omega_1 = \{*\}$ (the groups merged). We denote by Ω_{arr} the category with these two objects and with the merge $p: \omega_2 \rightarrow \omega_1$ as the only morphism different from the identities. We denote by Ω_{full} the full subcategory of FinSet on the same two objects. Its morphisms different from the identities are p , the injections $i_1, i_2: \omega_1 \rightarrow \omega_2$ (selection of a group), the swap σ (relabelling of the treatments) and the constants $c_k = i_k p$. Thus Ω_{full} has eight morphisms, with $p i_k = \text{id}$ and $\sigma i_1 = i_2$. We consider both categories because the merge alone does not detect the failure that we want to study (see Example 6.10(i)).

We identify $\omega_1 = \{*\}$ with $\{1\}$. Then i_1 is the insertion $\{1\} \subsetneq \{1, 2\}$, which corresponds by Lemma 5.6 to Tjur’s reduction to the first group, and $i_2 = \sigma i_1$ is that insertion followed by a relabelling. The reduction to the second group is the insertion $\{2\} \subsetneq \{1, 2\}$, which is represented by i_2 up to the same identification. Neither $\{1\}$ nor $\{2\}$ is an object of Ω_{full} , so that the insertions of Section 5 are represented in Ω_{full} and not contained in it, and “the insertion i_k ” below is always meant in this sense. By the cocycle identity of Lemma 6.3 we have $c(i_2) = A(\sigma)c(i_1) + c(\sigma)$, hence nothing depends on the identification that we choose.

We discretise the parameters. Let $\Theta(\omega) = M^\omega \times S$, where M and S are finite sets of admissible means and dispersions. Then $\Theta(p)(\mu, s) = ((\mu, \mu), s)$ is the diagonal, $\Theta(i_k)((\mu_1, \mu_2), s) = (\mu_k, s)$ is the k -th coordinate and $\Theta(\sigma)$ is the swap. Hence

$$A(\omega_1) = k^{M \times S}, \quad A(\omega_2) = k^{M^2 \times S}, \quad A(\psi) = \Theta(\psi)^*,$$

so that $A(p)$ is the restriction to the diagonal and $A(i_k)f$ is f as a function of the k -th coordinate. We take as g the *marginal*, or total, dispersion under balanced allocation,

$$g_{\omega_1}(\mu, s) = s, \quad g_{\omega_2}((\mu_1, \mu_2), s) = s + \frac{(\mu_1 - \mu_2)^2}{4}.$$

It is the analogue for the one-way layout of Tjur’s $\alpha + \beta \bar{x}$ of Example 5.12. Like that quantity, it is a parameter to which a term is added that depends on the groups included in the design. The target Λ is constant at the finite set $T := g_{\omega_2}(\Theta(\omega_2)) \subset \mathbb{Q}$ of the values of g , which contains $S = g_{\omega_1}(\Theta(\omega_1))$, with $\Lambda(\psi) = \text{id}_T$. Hence $B(\omega) = k^T$ for both objects and $B(\psi) = \text{id}$.

The Baues–Wirsching complex is finite dimensional. We use the normalised subcomplex, which has the same cohomology as the complex of [2, (1.4)], as for every cosimplicial abelian group. For both categories

$$C^0 = \text{Hom}_k(B(\omega_1), A(\omega_1)) \oplus \text{Hom}_k(B(\omega_2), A(\omega_2)), \quad (d^0 h)(\psi) = A(\psi)h_\omega - h_{\omega'} B(\psi).$$

The group C^1_{norm} has one component over Ω_{arr} , namely $\text{Hom}_k(B(\omega_2), A(\omega_1))$ at $\psi = p$, and six components over Ω_{full} , one for each of $p, i_1, i_2, \sigma, c_1, c_2$. The differential d^1 is the operator whose vanishing is the cocycle identity of Lemma 6.3, and $H^1 = \ker d^1 / \text{im } d^0$. Over Ω_{arr} no two morphisms different from the identities can be composed, hence $d^1 = 0$ and H^1 is the cokernel of d^0 . Thus everything is a finite computation with matrices once M, S and g are fixed.

What is computed

The computations are carried out by two scripts, in exact arithmetic over $\mathbb{Q}$.¹ Given one of the two categories, the finite sets M and S and the pair $(g_{\omega_1}, g_{\omega_2})$, the first script computes the defect c of (13), the dimensions of C^1, Z^1, B^1 and H^1 , a particular solution h of $c = -d^0 h$ with the corrected quantity, and the

¹Scripts and committed outputs are available at <https://github.com/Vaxx66/calamus>, in the release tagged v35, which also contains the version of this manuscript to which the numbers quoted below refer; a fresh run reproduces the committed outputs byte for byte. The repository describes the outputs of the scripts in detail.

partition of T of Proposition 6.9. It is run five times. Run 1 has Ω_{arr} , $M = \{0, 1\}$ and $S = \{1, 2\}$. Run 2 has Ω_{full} with the same M and S . Run 3 has Ω_{full} , $M = \{0, 1\}$ and $S = \{1, \frac{5}{4}\}$. Check A is run 1 with g_{ω_2} replaced by $g'_{\omega_2}((\mu_1, \mu_2), s) = s + \mu_1/4$, which does not agree with g_{ω_1} on the diagonal. Check B has Ω_{full} , $M = \{0, 1, 2\}$ and $S = \{1, 2\}$. The second script computes H^1 again in an independent way, through the reduced coefficient system of Example 6.12. All the numbers quoted below are read from the committed outputs. The statements which hold for all M and S are proved, and the runs are instances of them.

Results

We first identify the within-group dispersion as the correction of the marginal dispersion, which is canonical with respect to a given class.

Proposition 6.8. *In the setting of Example 6.7, over Ω_{full} and for any M and S , let*

$$C_1 := \{h \in C^0 : h_{\omega_1} = 0\}$$

be the class of corrections which leave the component at the merged design unchanged. Then $C_1 \cap \text{Nat}(B, A) = 0$, so that a correction in C_1 is unique when it exists. A deterministic raw family $g = (g_{\omega_1}, g_{\omega_2})$ with target constant at T has a correction in C_1 if and only if g_{ω_1} does not depend on the mean, i.e. if and only if $g_{\omega_1} \circ \Theta(i_1) = g_{\omega_1} \circ \Theta(i_2)$. In this case, the corrected family is deterministic, with component $g_{\omega_1} \circ \Theta(i_1)$ at ω_2 . For the marginal dispersion, the corrected component at ω_2 is the within-group dispersion $((\mu_1, \mu_2), s) \mapsto s$.

Proof. Since Λ is constant we have $B(\psi) = \text{id}$, and the equations $d^0 h = 0$ at i_1 and i_2 read $h_{\omega_2} = A(i_k)h_{\omega_1}$. Hence a natural 0-cochain is determined by its component at ω_1 , so that $C_1 \cap \text{Nat}(B, A) = 0$, and the uniqueness follows from Proposition 6.6(1). Let $h \in C_1$ be a correction. The natural family $g^* + h$ has component $g_{\omega_1}^*$ at ω_1 , hence its component at ω_2 is $A(i_1)g_{\omega_1}^* = A(i_2)g_{\omega_1}^*$. This is the stated condition, because $k^{(-)}$ is faithful and $A(i_k)g_{\omega_1}^* = (g_{\omega_1} \circ \Theta(i_k))^*$. Conversely, suppose that the condition holds and set $n_{\omega_1} = g_{\omega_1}^*$ and $n_{\omega_2} = A(i_1)g_{\omega_1}^*$. The equations $d^0 n = 0$ hold at the generators i_1, i_2, p, σ . Indeed they hold at i_1 by definition, at i_2 by the condition, at p because $A(p)A(i_1) = A(p i_1) = \text{id}$, and at σ because $A(\sigma)A(i_1) = A(\sigma i_1) = A(i_2)$. Hence n is natural, $h = n - g^*$ belongs to C_1 , and n is the dual of the deterministic family $(g_{\omega_1}, g_{\omega_1} \circ \Theta(i_1))$. For the marginal dispersion $g_{\omega_1}(\mu, s) = s$ does not depend on μ , and $g_{\omega_1} \circ \Theta(i_1)$ is $((\mu_1, \mu_2), s) \mapsto s$. $\square$

Proof of Theorem 1.5. The first two statements are Proposition 6.6(1) and (2). The statements on the one-way layout are Proposition 6.8. $\square$

We now consider the *recalibrations* of g , i.e. the families $r \circ g = (r_{\omega_1}g_{\omega_1}, r_{\omega_2}g_{\omega_2})$, where $r = (r_{\omega_1}, r_{\omega_2})$ is a pair of maps $T \rightarrow T$. A recalibration expresses the quantity on a different scale of values.

Proposition 6.9. *In the setting of Example 6.7, let $\sim$ be the equivalence relation on T generated by $s \sim s + (\mu_1 - \mu_2)^2/4$ for $s \in S$ and $\mu_1, \mu_2 \in M$. The recalibration $r \circ g$ is natural over Ω_{full} if and only if r_{ω_2} is constant on each class of $\sim$ and $r_{\omega_1} = r_{\omega_2}$ on S . Furthermore:*

1. *there is a natural recalibration with r_{ω_2} not constant if and only if $\sim$ has at least two classes;*
2. *if $|M| \geq 2$ then no natural recalibration has r_{ω_2} injective, i.e. the marginal dispersion cannot be expressed on another scale in a natural way without loss of information;*
3. *over Ω_{arr} the recalibration $r \circ g$ is natural if and only if $r_{\omega_1} = r_{\omega_2}$ on S .*

Proof. The square (11) at i_k reads $r_{\omega_1}(g_{\omega_1}(\Theta(i_k)\theta)) = r_{\omega_2}(g_{\omega_2}(\theta))$ for all $\theta = ((\mu_1, \mu_2), s)$, i.e. $r_{\omega_1}(s) = r_{\omega_2}(s + (\mu_1 - \mu_2)^2/4)$, and these are the same equations for $k = 1, 2$. The case $\mu_1 = \mu_2$ gives $r_{\omega_1} = r_{\omega_2}$ on S . Substituting, we obtain $r_{\omega_2}(s) = r_{\omega_2}(s + (\mu_1 - \mu_2)^2/4)$ for every generating pair of $\sim$, i.e. r_{ω_2} is constant on the classes, since every element of T is of the form $s + (\mu_1 - \mu_2)^2/4$. Conversely, these two conditions give the squares at i_1 and i_2 . The square at p is the case $\mu_1 = \mu_2$, the square at σ holds for every r because

g_{ω_2} is symmetric, and the squares at $c_k = i_k p$ follow by pasting. Part (1) holds because r_{ω_2} is not constant if and only if it separates two classes. For part (2), let $\mu_1 \neq \mu_2$ in M and $s \in S$. Then s and $s + (\mu_1 - \mu_2)^2/4$ are distinct elements of the same class, on which an injective r_{ω_2} cannot be constant. Part (3) is the square at p alone. $\square$

Example 6.10. (i) *The merge alone detects nothing (run 1 and check A).* Over Ω_{arr} the marginal dispersion is natural, i.e. its defect is zero, because the between-group term $(\mu_1 - \mu_2)^2/4$ vanishes on the diagonal $\Theta(p)$. This is a property of the quantity and not of the category. Indeed, in check A the same category is given a quantity which does not agree with g_{ω_1} on the diagonal. The script finds a defect different from zero at p and corrects it, and the particular solution leaves g'_{ω_2} unchanged and replaces the component at the merged design with $(\mu, s) \mapsto s + \mu/4$. The cohomology detects nothing either, for every M and S . Indeed $B(p) = \text{id}$ gives that d^0 is surjective onto $C^1_{\text{norm}} = \text{Hom}_k(B(\omega_2), A(\omega_1))$, and $d^1 = 0$. In run 1 we have $\dim C^1 = \dim Z^1 = \dim B^1 = 16$ and $H^1 = 0$.

(ii) *The quantity is not meaningful, and a deterministic correction is found (run 2).* Over Ω_{full} with $M = \{0, 1\}$ and $S = \{1, 2\}$, so that $T = \{1, \frac{5}{4}, 2, \frac{9}{4}\}$, the defect is different from zero on $\{i_1, i_2, c_1, c_2\}$ and it is zero on p and σ . The cocycle identity holds entry by entry on all the 22 composable pairs. Thus the failure is detected by the selections i_1, i_2 of a group and not by the merge, i.e. it is detected by the criterion of Tjur, which quantifies over the removals of sample points only. The defect at the insertion i_1 is a violation of his condition at the reduction to the first group (Lemma 5.6). Here $\dim C^1 = 176$, $\dim Z^1 = \dim B^1 = 40$ and $H^1 = 0$. The group vanishes although the defect does not, in agreement with Proposition 6.4(2), and its vanishing gives no information on the quantity. Solving $c = -d^0 h$ over the whole group C^0 , which is always possible by the same part, we find a corrected family which is again deterministic. The solution obtained by setting to zero the free parameters of the row-reduced system has the *within-group* dispersion $((\mu_1, \mu_2), s) \mapsto s$ as component at ω_2 , i.e. the between-group term is removed. The set of all the solutions is the affine space $-g^* + \text{Nat}(B, A)$, which contains other deterministic elements, so that this solution is not canonical among all the corrections. It is canonical with respect to the class C_1 of Proposition 6.8. Indeed, a natural family over Ω_{full} is determined by its component at ω_1 , so that a corrected family whose component at ω_2 is the within-group dispersion leaves g_{ω_1} unchanged, i.e. the correction belongs to C_1 . Thus the statistically sensible correction is the unique one which does not change the design on which g is already a within-group dispersion.

(iii) *An obstruction different from zero with respect to an admissible class (runs 2, 3 and check B).* We take as admissible the recalibrations $r \circ g$ with r_{ω_2} not constant, so that the quantity remains a non-trivial function of the observed marginal dispersion. The statistically natural class is the smaller one of the injective recalibrations, which preserve the information, and by Proposition 6.9(2) it contains no natural element for any grid with $|M| \geq 2$. We keep the larger class because it is the weakest one which can carry an obstruction, so that an obstruction with respect to it is not a consequence of injectivity. By Proposition 6.9, naturality forces r_{ω_2} to be constant on the classes of the partition of T generated by $s \sim s + (\mu_1 - \mu_2)^2/4$, and the existence of a natural recalibration is decided by computing this partition. We say that the design is *scale-confounded* if some marginal value $s + (\mu_1 - \mu_2)^2/4$ with $\mu_1 \neq \mu_2$ is itself an admissible within-group dispersion, i.e. it belongs to S . With $M = \{0, 1\}$ and $S = \{1, \frac{5}{4}\}$ (run 3, where $T = \{1, \frac{5}{4}, \frac{3}{2}\}$), or with $S = \{1, 2\}$ and three levels of the mean $M = \{0, 1, 2\}$ (check B), the partition has a single class. Hence every natural recalibration is constant, and the obstruction with respect to this class is not zero, i.e. no recalibration of the marginal dispersion which carries information is natural. With $M = \{0, 1\}$ and $S = \{1, 2\}$ as in (ii) the partition has the two classes $\{1, \frac{5}{4}\}$ and $\{2, \frac{9}{4}\}$ (run 2), and there is a natural recalibration which is not constant, but no bijective one. In runs 2, 3 and in check B the unrestricted correction of (ii) gives the within-group dispersion, i.e. the correction in C_1 of Proposition 6.8.

The finite runs discretise a statement which holds in the Gaussian layout itself, where it is simpler.

Proposition 6.11 (Continuous case). *Let $\Theta(\omega) = \mathbb{R}^\omega \times \mathbb{R}_{>0}$ as in Example 2.7, let g be the marginal dispersion, i.e. $g_{\omega_1}(\mu, s) = s$ and $g_{\omega_2}((\mu_1, \mu_2), s) = s + (\mu_1 - \mu_2)^2/4$, with target constant at $T = \mathbb{R}_{>0} = g_{\omega_2}(\Theta(\omega_2))$, and let $r_{\omega_1}, r_{\omega_2} : T \rightarrow T$ be any maps. If $r \circ g$ satisfies Tjur's condition at the reduction to the first group, i.e. if the square (11) at i_1 commutes, then $r_{\omega_1} = r_{\omega_2}$ is constant. Hence no recalibration of the marginal dispersion, which is not constant, is Tjur-natural, and a fortiori none is meaningful.*

Proof. The square at i_1 reads $r_{\omega_1}(s) = r_{\omega_2}(s + (\mu_1 - \mu_2)^2/4)$ for all $s > 0$ and all real μ_1, μ_2 . For $\mu_1 = \mu_2$ it gives $r_{\omega_1} = r_{\omega_2}$. For $0 < s < s'$, choosing $\mu_1 - \mu_2 = 2\sqrt{s' - s}$ we obtain $r_{\omega_2}(s) = r_{\omega_2}(s')$. $\square$

Note that no measurability is used and that the class of the recalibrations is not restricted in advance. In the continuous case every admissible marginal value is confounded with every within-group dispersion. The partition of T in the finite runs discretises this argument. Its two classes in run 2 depend on the grid, where the only available step is $(1 - 0)^2/4 = \frac{1}{4}$ and $\frac{5}{4} \notin S$ interrupts the chain from 1 to 2. A third level of the mean (check B) or a scale-confounded S (run 3) restores the connection. Thus the finite computation is an exact certificate for the given grid, and the continuous statement says what the grids approximate.

Example 6.12. We compute $H^1(\Omega; \text{Hom}_k(B, A))$, see Proposition 6.4(3). For Ω_{arr} it is zero in every run, as it must be by Example 6.10(i). For Ω_{full} it is zero with two levels of the mean (runs 2 and 3), and it is not zero with three levels. With $M = \{0, 1, 2\}$, $S = \{1, 2\}$ and Λ constant at T as above, so that $|T| = 5$, check B gives $\dim C^1 = 480$, $\dim Z^1 = 120$ and $\dim B^1 = 110$, hence $\dim_k H^1(\Omega_{\text{full}}; \text{Hom}_k(B, A)) = 10 = 2|T|$. The group depends on Λ as well as on Ω . Indeed, when B is constant with $B(\psi) = \text{id}$ it is the direct sum of $|T|$ copies of the cohomology with coefficients $\psi \mapsto A(\text{cod } \psi)$. This coefficient system is the pullback along $F\Omega \rightarrow \Omega$ of the Ω -module A [2, (1.18)(2)], hence by [2, (8.5)(B)]

$$H^1(\Omega; \text{Hom}_k(B, A)) \cong (\varprojlim_{\Omega}^{(1)} A)^{\oplus |T|},$$

i.e. $|T|$ copies of the first derived limit of the parameter diagram, and Λ appears only through $|T|$. The second script computes the reduced group directly and confirms this factorisation. Its dimension is 2 for $|M| = 3$ (and $2 \cdot 5 = 10$) and 6 for $|M| = 4$ (where $|T| = 7$, and $6 \cdot 7 = 42$), so that the group grows with the number of the levels of the mean. Its algebraic meaning is the usual one for a first derived limit. When $B = A$ the cocycle identity of Lemma 6.3 says that $A(\psi) + \varepsilon c(\psi)$ is functorial modulo ε^2 , and $d^0 h$ is the deformation induced by the change of coordinates $1 + \varepsilon h_\omega$ at each object. Hence H^1 is the space of the first-order deformations of the parameter diagram, as a diagram of vector spaces, modulo these changes of coordinates. We do not know its statistical meaning.

7 The ambient category for the Bayesian shape

In Sections 8 and 9 we use states, conditionals and almost sure equality, which do not appear in the previous sections. They are synthetic notions of Fritz [10] and of Cho and Jacobs [7], and we recall them here with the references to the two sources.

7.1 States and the category BorelStoch

Let $(\mathbb{C}, \otimes, I)$ be a Markov category. The unit I is terminal, and a *state* on X is a morphism $I \rightsquigarrow X$. In FinStoch a state is a probability vector [10, Ex. 2.5], and in Stoch it is a probability measure. The deterministic morphisms are those of Definition 3.2 [10, Def. 10.1].

The category Stoch is not suitable for our purposes, for two reasons. Its deterministic morphisms need not come from Meas [10, Ex. 10.4], and conditionals need not exist [10, Ex. 11.3]. Both problems disappear in BorelStoch , the full subcategory of Stoch of the *standard Borel* spaces, i.e. of the measurable spaces which are isomorphic to a Polish space with its Borel σ -algebra, equivalently to a Borel subset of $\mathbb{R}$ [10, p. 21], [7, p. 12]. It contains I , and it is closed under $\otimes$, hence it is a Markov category, and it is the Kleisli category of the Giry monad restricted to the standard Borel spaces [7, p. 12]. Its deterministic morphisms are the measurable maps [10, Ex. 10.5], so that δ restricted to the standard Borel spaces is fully faithful onto them. Note that Kolmogorov products exist in BorelStoch only for countable index sets [10, p. 21], [24, Ex. 3.6] (see Remark 8.7).

Throughout Sections 8 and 9 the ambient category is BorelStoch , and we use Theorem 4.4 only for the models whose parameter objects $\Theta_{\text{Meas}}(\omega)$ and sample spaces $\Gamma(v, u)$ are standard Borel, so that $\mathbf{L}$, $\mathbf{R}$ and the components $\mathbf{P}_{v,d}$ belong to BorelStoch . This holds in the finite one-way scheme of Example 6.7, where every object is finite, and in Example 2.12, where $\Theta(\omega) = \omega^* \times \mathbb{R}_{>0}$ and $\Gamma(\mathbb{R}, u) = \mathbb{R}^u$. All the statements

below which do not use conditioning, i.e. all except the posterior of Definition 8.1 and Proposition 9.2, hold in Stoch and, in the finite scheme, in FinStoch. Sections 8 and 9 do not depend on the appendix.

7.2 Conditionals and Bayesian inversion

Let $\nu: I \rightsquigarrow X \otimes Y$ be a state, which we think of as a joint distribution, and let ν_X and ν_Y be its marginals, i.e. its compositions with the discard maps of Y and of X respectively. A *conditional* of ν given X is a morphism $\nu|_X: X \rightsquigarrow Y$ such that

$$\nu = (\text{id}_X \otimes \nu|_X) \circ \text{copy}_X \circ \nu_X, \quad (16)$$

and a conditional of ν given Y is a morphism $\nu|_Y: Y \rightsquigarrow X$ such that $\nu = (\nu|_Y \otimes \text{id}_Y) \circ \text{copy}_Y \circ \nu_Y$. The latter is a conditional given the first factor of the composition of ν with the symmetry $X \otimes Y \cong Y \otimes X$. Cho and Jacobs call these morphisms *disintegrations* of ν [7, Def. 3.5], and a Markov category *has conditional distributions* if every state on a product has them [10, Def. 11.1]. In Stoch the two equations read

$$\nu(S \times T) = \int_S \nu|_X(T | x) \nu_X(dx) = \int_T \nu|_Y(S | y) \nu_Y(dy) \quad (S \in \Sigma_X, T \in \Sigma_Y),$$

i.e. $\nu|_X$ and $\nu|_Y$ are regular conditional probabilities of ν given the first and the second coordinate [10, Ex. 11.3, eq. (11.3)], [7, Ex. 3.7]. Such kernels need not exist for arbitrary measurable spaces, and $\nu|_X$ exists when Y is standard Borel [7, Thm. 3.11]. Hence BorelStoch has conditional distributions [10, Ex. 11.3], [7, Cor. 3.12], while Stoch does not.

Let $\alpha: I \rightsquigarrow X$ be a state. Two morphisms $f, g: X \rightsquigarrow Y$ are called *α -a.s. equal* if $(\text{id}_X \otimes f) \circ \text{copy}_X \circ \alpha = (\text{id}_X \otimes g) \circ \text{copy}_X \circ \alpha$ [7, Def. 5.1], [10, Def. 13.1]. In Stoch this means that $f(T | -)$ and $g(T | -)$ agree α -almost everywhere for every $T \in \Sigma_Y$ [7, Prop. 5.3], [10, Ex. 13.3]. By definition, two conditionals of ν given X are ν_X -a.s. equal, and by symmetry two conditionals given Y are ν_Y -a.s. equal [7, Prop. 5.2], [10, Prop. 13.7].

Let now $\pi: I \rightsquigarrow X$ be a state and let $k: X \rightsquigarrow Y$ be a morphism. We set

$$\nu := (\text{id}_X \otimes k) \circ \text{copy}_X \circ \pi: I \rightsquigarrow X \otimes Y \quad \text{and} \quad m := k \circ \pi: I \rightsquigarrow Y.$$

Then $\nu_X = \pi$, $\nu_Y = m$ and k is a conditional of ν given X . A conditional of ν given Y is called a *Bayesian inversion* of k with respect to π , and it is denoted by $k_\pi^\dagger: Y \rightsquigarrow X$. Thus $k_\pi^\dagger$ is a morphism such that

$$(\text{id}_X \otimes k) \circ \text{copy}_X \circ \pi = (k_\pi^\dagger \otimes \text{id}_Y) \circ \text{copy}_Y \circ m \quad (17)$$

(see [7, eq. (5)] and [10, eq. (13.2)]). In statistical terms, X is the parameter space, Y is the sample space, π is the prior, k is the model, ν is their joint distribution, m is the prior predictive distribution and $k_\pi^\dagger$ is the posterior. In Stoch equation (17) is Bayes' theorem in the form

$$\int_S k(T | x) \pi(dx) = \int_T k_\pi^\dagger(S | y) m(dy) \quad (S \in \Sigma_X, T \in \Sigma_Y)$$

[7, Ex. 3.9], where both sides are the probability that the parameter belongs to S and the observation to T . Since a Bayesian inversion is a conditional, it exists in BorelStoch, and it is unique up to m -a.s. equality. We write $k^\dagger$ when the prior is fixed.

8 Priors as states, coherence as reduction-naturality

In Sections 4–6 the parameter side of a bridged model is deterministic, and meaningfulness is a condition on the functionals of the parameter. In this section we consider the priors, which are families of states on the parameter objects, and we prove parts (1) and (2) of Theorem 1.7. We shall see that the two criteria of Section 5 play different roles for them. The definitions and the results up to Remark 8.3 hold for every bridged model which satisfies the convention of Section 7. We then apply them to the finite one-way scheme, since the priors given by the rules of Jeffreys for the Gaussian one-way layout are improper; hence they are not states (see Proposition 8.5), and in Section 9 to the Gaussian linear model.

8.1 Conventions

Throughout this section, a response scale $v \in \text{ob } V$ and a unit object $u \in \text{ob } U$ are fixed, and only the covariate space varies. Let $D_u \subseteq D$ be the subcategory of the designs (u, ω, x) and of the morphisms (id_u, ψ) , for which (1) reads $x' = J\psi \circ x$. Thus the target design is determined by x and ψ . Suppose that every ω admits a design map from u , as in Example 2.7. Then a family indexed by $\text{ob } \Omega$ is natural over Ω if and only if its pullback to D_u is natural, and for this reason we state the conditions over Ω although the models are defined over D . On D_u^{op} the sample side of the bridged model is constant, i.e. $\mathbf{R}(u, \omega, x) = \Gamma(v, u)$ and $\mathbf{R}((\text{id}_u, \psi)^{\text{op}}) = \text{id}$, and we write $\mathbf{R}(\omega) = \Gamma(v, u)$ and $\mathbf{R}(\psi) = \text{id}$. The components of the model depend on the design map, and we write

$$\mathbf{P}_\omega := \mathbf{P}_{v,d} \quad \text{for a design } d = (u, \omega, x) \in \text{ob } D_u \text{ over } \omega.$$

Each identity below is stated for every such d , with target design $d' = (u, \omega', J\psi \circ x)$. We do not choose one design for each covariate space in a coherent way, and in general this is not possible. Indeed in the one-way layout the two selections $i_1, i_2: \{*\} \rightarrow \{1, 2\}$ would give $i_1 \circ x = i_2 \circ x$, which fails for $u \neq \emptyset$. With these conventions, the naturality of the bridged model $\mathbf{P}$ of Theorem 4.4 at $\psi: \omega \rightarrow \omega'$ is the square (6) at (id_u, ψ) , i.e.

$$\mathbf{R}(\psi) \circ \mathbf{P}_{\omega'} = \mathbf{P}_\omega \circ \delta\Theta_{\text{Meas}}(\psi) \quad \text{in Stoch.} \quad (18)$$

By (9) it is condition (N- Ω) of Section 2.3, written for the kernels $\mathbf{P}_{v,d}$ instead of the maps $P_{v,d}$. We keep the factor $\mathbf{R}(\psi)$, which is an identity here, because in this form the proofs below hold over D , where $\mathbf{R}$ acts by marginalisation. The quantifier classes $\Omega_{\text{red}} \subseteq \Omega_{\text{inj}} \subseteq \Omega$ are those of Section 5.

8.2 Priors and coherence

Definition 8.1 (Prior; coherence). Let $\mathbf{P}$ be a bridged model. A *prior* for $\mathbf{P}$ is a family of states

$$\pi_\omega: I \rightsquigarrow \Theta_{\text{Meas}}(\omega), \quad \omega \in \text{ob } \Omega,$$

i.e. of probability measures on the parameter objects. The associated *Bayesian model* at ω is the joint state $(\text{id} \otimes \mathbf{P}_\omega) \circ \text{copy} \circ \pi_\omega: I \rightsquigarrow \Theta_{\text{Meas}}(\omega) \otimes \mathbf{R}(\omega)$, and the *prior predictive* is its marginal on $\mathbf{R}(\omega)$

$$m_\omega := \mathbf{P}_\omega \circ \pi_\omega: I \rightsquigarrow \mathbf{R}(\omega).$$

The *posterior* is the Bayesian inversion $\mathbf{P}_\omega^\dagger: \mathbf{R}(\omega) \rightsquigarrow \Theta_{\text{Meas}}(\omega)$ of $\mathbf{P}_\omega$ with respect to π_ω , i.e. a conditional of the Bayesian model given $\mathbf{R}(\omega)$ (see (17)). It exists in BorelStoch , and it is unique up to m_ω -a.s. equality.

Let $Q \subseteq \Omega$ be a subcategory which contains all the objects. The prior is called *Q-coherent* if the family $(\pi_\omega)_\omega$ is *Q-coherent* in the sense of Definition 5.1 with $F = \Delta_I$ and $G = \delta\Theta_{\text{Meas}}$, where $\Delta_I: \Omega^{\text{op}} \rightarrow \text{BorelStoch}$ is the constant functor at the monoidal unit I . Explicitly, this means that

$$\pi_\omega = \delta\Theta_{\text{Meas}}(\psi) \circ \pi_{\omega'} = \Theta_{\text{Meas}}(\psi)_* \pi_{\omega'} \quad (19)$$

for every $\psi: \omega \rightarrow \omega'$ in Q . We say that the prior is *coherent at ψ* if (19) holds at ψ , so that *Q-coherence* is coherence at every morphism of Q . We say that the prior is *coherent* if it is Ω_{red} -coherent, i.e. natural along the insertions, which is the class of Tjur. We say that it is *relabelling-coherent* if it is Ω_{inj} -coherent, i.e. natural along all the injections, which is the class of McCullagh as stated. Since $\Omega_{\text{red}} \subseteq \Omega_{\text{inj}}$, the second condition implies the first one.

In words, a prior is coherent if the prior of a smaller design is the marginal of the prior of every larger design which contains it.

Proposition 8.2 (Structure of coherence). *Let $\mathbf{P}$ be a bridged model, let π be a prior and let $Q \subseteq \Omega$ be a subcategory which contains all the objects.*

1. If (19) holds at $\psi: \omega \rightarrow \omega'$ then π_ω vanishes on every measurable subset of $\Theta_{\text{Meas}}(\omega)$ which is disjoint from $\text{im } \Theta_{\text{Meas}}(\psi)$. If the image is measurable, this means that $\pi_\omega(\text{im } \Theta_{\text{Meas}}(\psi)) = 1$, and in the finite case that $\text{supp } \pi_\omega \subseteq \text{im } \Theta_{\text{Meas}}(\psi)$.
2. Let $\psi_1, \psi_2: \omega \rightarrow \omega'$ be two morphisms of $\mathbf{Q}$. If π is $\mathbf{Q}$ -coherent then $\Theta_{\text{Meas}}(\psi_1)_* \pi_{\omega'} = \Theta_{\text{Meas}}(\psi_2)_* \pi_{\omega'}$. In particular, if π is Ω_{inj} -coherent then π_ω is invariant under $\Theta_{\text{Meas}}(\text{Aut } \omega)$ for every ω , while Ω_{red} -coherence imposes no symmetry on any component.
3. If π is $\mathbf{Q}$ -coherent then $m_\omega = \mathbf{R}(\psi) \circ m_{\omega'}$ for every $\psi: \omega \rightarrow \omega'$ in $\mathbf{Q}$ and every design d over ω , where m_ω is computed at d and $m_{\omega'}$ at the target design d' . Thus the prior predictives, indexed by the designs of $\mathbf{D}_u$, form a natural transformation $\Delta_I \Rightarrow \mathbf{R}$ over the morphisms (id_u, ψ) with ψ in $\mathbf{Q}$, and we say that m is natural over $\mathbf{Q}$.
4. Suppose that, for each design d over ω , the map $\alpha \mapsto \mathbf{P}_\omega \circ \alpha$ from the states on $\Theta_{\text{Meas}}(\omega)$ to the states on $\mathbf{R}(\omega)$ is injective on a class $\mathbf{S}_\omega$ of states which contains π_ω and satisfies $\Theta_{\text{Meas}}(\psi)_* \mathbf{S}_{\omega'} \subseteq \mathbf{S}_\omega$ for every $\psi: \omega \rightarrow \omega'$ in $\mathbf{Q}$, for example on all the states, i.e. when the model is identifiable in mixtures. If m is natural over $\mathbf{Q}$ then π is $\mathbf{Q}$ -coherent. In FinStoch the injectivity on all the states means that the rows of the stochastic matrix $\mathbf{P}_\omega$, indexed by $\Theta(\omega)$, are linearly independent.

Proof. (1) Let f be a measurable map and let A be a measurable set which is disjoint from $\text{im } f$. Then $(f_* \nu)(A) = \nu(f^{-1}A) = \nu(\emptyset) = 0$. Note that the image need not be measurable, and for this reason the statement is given on the sets which are disjoint from it.

(2) The first statement follows from (19) applied to ψ_1 and to ψ_2 . For the automorphisms take $\psi_1 = \sigma$ and $\psi_2 = \text{id}_\omega$. The morphisms of Ω_{red} are inclusions, and an inclusion is determined by its source and its target, so that Ω_{red} has no pairs of distinct parallel morphisms.

(3) Let ψ be a morphism of $\mathbf{Q}$. By the naturality of $\mathbf{P}$ in the form (18) we have

$$\mathbf{R}(\psi) \circ m_{\omega'} = \mathbf{R}(\psi) \circ \mathbf{P}_{\omega'} \circ \pi_{\omega'} = \mathbf{P}_\omega \circ \delta \Theta_{\text{Meas}}(\psi) \circ \pi_{\omega'} = \mathbf{P}_\omega \circ \pi_\omega = m_\omega.$$

(4) The same chain of equalities gives $\mathbf{P}_\omega \circ (\Theta_{\text{Meas}}(\psi)_* \pi_{\omega'}) = \mathbf{P}_\omega \circ \pi_\omega$. Both states belong to $\mathbf{S}_\omega$, and the result follows by the injectivity hypothesis. For the statement on FinStoch , write the states as row vectors, so that $\mathbf{P}_\omega \circ \alpha = \alpha \mathbf{P}_\omega$. If $\nu \mathbf{P}_\omega = 0$ then $\sum_\theta \nu_\theta = \sum_y (\nu \mathbf{P}_\omega)_y = 0$, because the rows sum to one. Hence the kernel of $\nu \mapsto \nu \mathbf{P}_\omega$ is contained in the hyperplane of the vectors with zero sum, which is spanned by the differences of probability vectors, and the map is injective on the states if and only if it is injective, i.e. if and only if the rows are linearly independent. $\square$

Remark 8.3. Consider the merge $p: \{1, 2\} \rightarrow \{*\}$ of the one-way layout, for which $\Theta(p)$ is the diagonal, both in the Gaussian model of Example 2.7 and in the finite scheme of Example 6.7. The diagonal is measurable, hence by part (1), if (19) holds at p then $\pi_{\{1,2\}}$ is concentrated on it, so that the prior for two groups carries no more information than the prior for one group. For this reason (19) is imposed over Ω_{red} or over Ω_{inj} and not over Ω . Consider now part (2) in the one-way layout, with $\omega = \{1, \dots, n\}$ and $\Theta(\sigma)(\mu, s) = (\mu \circ \sigma, s)$ for a permutation σ . The invariance under $\Theta_{\text{Meas}}(\text{Aut } \omega)$ is the invariance of π_ω under all the permutations of the coordinates of the mean, jointly with the dispersion, i.e. the finite counterpart of the exchangeability of [11, Def. 4.1], which is stated there for a state on a countable Kolmogorov power (see Remark 8.7). At the two injections $\{*\} \rightarrow \{1, 2\}$ part (2) says that the two coordinate marginals of $\pi_{\{1,2\}}$ agree, and at the relabelling $\{1\} \rightarrow \{2\}$, composed with the inclusions into $\{1, 2\}$, it says that the two priors for one group coincide. Finally, note that the hypothesis of part (4) is stronger than the injectivity of $\theta \mapsto \mathbf{P}_\omega(\theta)$.

8.3 The finite one-way scheme

Proposition 8.4 (The four strata of the two-object scheme). *In the finite one-way scheme of Example 6.7 we have $\Theta(\omega_n) = M^n \times S$ with $\omega_1 = \{*\}$ and $\omega_2 = \{1, 2\}$, and the morphisms of Ω_{full} are the merge p , the*

selections i_1, i_2 , the swap σ and the constants $c_k = i_k p$. We identify ω_1 with $\{1\}$. Then the traces on Ω_{full} of the classes of Section 5 are

$$\Omega_{\text{red}} = \langle i_1 \rangle \subsetneq \Omega_{\text{sel}} := \langle i_1, i_2 \rangle \subsetneq \Omega_{\text{inj}} = \langle i_1, i_2, \sigma \rangle \subsetneq \Omega_{\text{full}},$$

where Ω_{sel} is not one of the classes of Section 5. It comes from the identification of the two designs with one group $\{1\}$ and $\{2\}$ with the single object ω_1 , i.e. from the relabelling $\{1\} \rightarrow \{2\}$ imposed without the relabelling σ of the design with two groups. For each of the four classes, the set of the $\mathbb{Q}$ -coherent priors is a polytope, of affine dimension

$$\begin{aligned} \Omega_{\text{red}} : & \quad |M|^2 |S| - 1 & \quad (\text{no constraint on } \pi_{\omega_2}), \\ \Omega_{\text{sel}} : & \quad |M|^2 |S| - 1 - |S|(|M| - 1) & \quad (\text{the two coordinate marginals of } \pi_{\omega_2} \text{ agree}), \\ \Omega_{\text{inj}} : & \quad \frac{1}{2} |M|(|M| + 1) |S| - 1 & \quad (\pi_{\omega_2} \text{ symmetric under the swap}), \\ \Omega_{\text{full}} : & \quad |M| |S| - 1 & \quad (\pi_{\omega_2} \text{ supported on the diagonal}). \end{aligned}$$

The second and the third dimensions coincide if and only if $|M| \leq 2$, and their difference is $|S| \binom{|M|-1}{2}$. Thus for $|M| = 2$ the equality of the two marginals implies the symmetry, and for $|M| \geq 3$ it does not.

Proof. In all the cases π_{ω_1} is determined by π_{ω_2} through (19) at i_1 , so that the free variable is π_{ω_2} , a probability vector on $M^2 \times S$. For the first three classes the uniform vector on $M^2 \times S$ satisfies the constraints below and has positive coordinates, so that it lies in the relative interior of the polytope, and the affine dimension of the polytope is the one of the affine space of solutions. For Ω_{full} the polytope is the image of the simplex on $M \times S$ under the injective affine map $\pi_{\omega_1} \mapsto (\pi_{\omega_1}, \Theta(p)_* \pi_{\omega_1})$, and its dimension is the one of the simplex. Over Ω_{red} there is no further condition. Over Ω_{sel} the only further condition is Proposition 8.2(2) at i_1, i_2 , i.e. that the two coordinate marginals of π_{ω_2} agree. The difference of the two marginalisation maps $\mathbb{Q}^{M^2 \times S} \rightarrow \mathbb{Q}^{M \times S}$ takes its values in the subspace of the vectors with zero sum over M for each s , and it is onto it, hence its rank is $|S|(|M| - 1)$. Its row space does not contain the normalisation functional, since every row functional vanishes on the diagonal entries. This gives the second dimension. Over Ω_{inj} the condition at σ is the symmetry of π_{ω_2} under the swap, which implies the equality of the marginals, and the symmetric probability vectors are parametrised by the $\frac{1}{2} |M|(|M| + 1) |S|$ orbits of the swap on $M^2 \times S$. Over Ω_{full} the condition at p confines π_{ω_2} to the diagonal, which has $|M| |S|$ points, and then the conditions at i_k, σ and c_k hold. Finally, the difference of the second and the third dimensions is $|S|(|M|^2 - |M| + 1 - \frac{1}{2} |M|(|M| + 1)) = |S| \frac{1}{2} (|M| - 1)(|M| - 2)$. $\square$

The four formulas have been verified exactly over $\mathbb{Q}$ for $|M| \leq 4$ and $|S| \leq 3$ by the script `oneway_prior_coherence.py` (see the end of this section).

Proposition 8.5 (Jeffreys' rule is not coherent, the independence prior is: finite analogue). *Consider the Gaussian one-way layout, with $\theta_\omega = (\mu \in \mathbb{R}^\omega, \sigma^2 > 0)$, at a design in which every treatment group is non-empty, so that the Fisher information is nonsingular. Jeffreys' general rule $\pi \propto \sqrt{\det I(\theta)}$ [17], [19, p. 1345, eq. (1)], gives*

$$\pi_\omega^J(d\mu, d\sigma^2) \propto (\sigma^2)^{-(|\omega|+2)/2} d\mu d\sigma^2,$$

whose exponent depends on the number $|\omega|$ of the mean parameters, hence on the design. The prior obtained by treating the location parameters separately, $\pi_\omega^J \propto (\sigma^2)^{-1}$, which is Jeffreys' own recommendation for the location–scale families [18, pp. 182–183], [19, p. 1345, eq. (2)] and is called the independent Jeffreys prior in the later literature [8, pp. 631–632], has an exponent which does not depend on $|\omega|$. Consider the proper finite analogue, where S is a finite set of squares of rational numbers and π_{ω_n} is uniform in μ with weight $s^{-(n+2)/2}$ in s , normalised.

1. A product rule $\pi_\omega = \text{unif}(M^\omega) \otimes \rho_\omega$ is coherent if and only if ρ_ω does not depend on ω . Hence the analogue of π^J is coherent and the analogue of π^J is not.
2. The defect of π^J is contained in the marginal of the dispersion. Its total variation along each of the two selections of a group is equal to the total variation between the normalisations of $s^{-3/2}$ and s^{-2}

on S , and it does not depend on M . For $M = \{0, 1\}$ and $S = \{1, 4\}$ the two marginals are $(\frac{8}{9}, \frac{1}{9})$ and $(\frac{16}{17}, \frac{1}{17})$, and the defect is $\frac{8}{153}$. Let ρ_{ω_n} be the normalised marginal of the dispersion of $\pi_{\omega_n}^J$. The defect along the merge is

$$\frac{1}{2} \left[(1 - |M|^{-1}) + \sum_{s \in S} |M|^{-1} \rho_{\omega_2}(s) - \rho_{\omega_1}(s) \right] \geq 1 - |M|^{-1},$$

where $1 - |M|^{-1}$ is the mass of $\pi_{\omega_2}^J$ outside the diagonal. The equality holds if and only if $\rho_{\omega_1}(s) \geq |M|^{-1} \rho_{\omega_2}(s)$ for every s , i.e. if and only if $Z_1/Z_2 \leq |M| s_{\min}^{1/2}$, where $Z_n = \sum_s s^{-(n+2)/2}$. The bound is attained in every small case, for example in the one above, but it is not an identity. For $|M| = 2$ and $S = \{1/4\} \cup \{(k/100)^2 : 105 \leq k \leq 229\}$ the defect along the merge is $0.5078 \dots > \frac{1}{2}$.

Part (1) and the numerical content of part (2), i.e. the formula for the merge, the cases where the bound is attained and the case where it is strict, are established for the finite analogue by exact computation over $\mathbb{Q}$ (script `oneway_prior_coherence.py`). The statement on π^J and π^{ij} in the continuous case is not a statement on Stoch . Indeed, both priors are improper; hence they are not states, i.e. they are not morphisms $I \rightsquigarrow \Theta_{\text{Meas}}(\omega)$. In the continuous case we only assert the computation of the exponent, which is classical (see Remark 8.6).

The dependence of the exponent on the design is Jeffreys' own objection to his general rule. Applied to k unknown means with a common variance, the general rule gives marginal t posteriors whose degrees of freedom depend on the total number of observations only, whatever k is, and he found this unacceptable [18, p. 182] (as reported in [19, p. 1345]). His remedy was to treat the location parameters separately, which gives π^{ij} . Part (1) says that the dependence on the design is a failure of naturality along the insertions, and that his remedy is the unique product rule which restores it. This agrees with, but is not the same as, the preference in the objective Bayes literature for the reference priors over the multivariate Jeffreys rule when nuisance parameters are present. For the normal location–scale family the reference prior for μ is σ^{-1} and not Jeffreys' σ^{-2} [4, p. 119], the multiparameter reference process is sequential in order to avoid the marginalisation paradoxes [3, pp. 905–906], and the multidimensional Jeffreys rule is recorded as a source of incoherence and of paradoxes [8, p. 631]. We do not claim that coherence characterises the reference priors.

Remark 8.6 (Open problems). The priors π^J and π^{ij} are improper, and to consider them one has to replace BorelStoch in Definition 8.1 and in Proposition 8.2 with a codomain which admits unnormalised states, for example the measurable spaces with the s -finite kernels, or the Kleisli category of the measure monad. We do not know whether Proposition 8.2 holds in such a codomain, whether Proposition 8.5 then holds for π^J and π^{ij} as stated in the continuous case, and whether coherence over Ω_{red} singles out a recognisable class of objective priors beyond the product rules of Proposition 8.5(1). We expect only the first question to be routine.

In the marginalisation paradoxes of Dawid, Stone and Zidek [9, §1 and Thm. 2.1, p. 198] the marginal posterior of a parameter ζ depends on the data only through a statistic z whose sampling density depends only on ζ , and it is not proportional to $\pi(d\zeta)f(z | \zeta)$ for any prior π . These are failures of commutation between the Bayesian inversion and the marginalisation of the parameter induced by a design reduction, i.e. of squares of the form (11) whose vertical arrows are the inversions $\mathbf{P}_{\omega}^{\dagger}$. We conjecture that in a codomain which admits unnormalised states, they are the failures of naturality of the family $(\mathbf{P}_{\omega}^{\dagger})_{\omega}$ over Ω_{red} . A stronger statement is not possible, for two reasons. First, Dawid, Stone and Zidek prove that the paradoxes cannot arise from proper priors [9, p. 189 and p. 194, eqs. (1.20)–(1.23)], while the incoherence of Proposition 8.5 already occurs for proper priors, so that the two phenomena are not the same. Their classification of the problems as reducible or irreducible and of the priors as paradoxical or paradox-free [9, p. 208] concerns the priors for a fixed problem, while coherence compares the priors across the designs, and it is only conjectured that the reference priors avoid the paradoxes [4, p. 123]. Second, the conjecture cannot be stated in Stoch , since improper priors are not states. Thus we do not claim that the present framework explains the marginalisation paradoxes.

Remark 8.7 (Kolmogorov consistency and exchangeability). Condition (19) over Ω_{red} says that $(\pi_{\omega})_{\omega}$ is a compatible family over the diagram of the reductions, i.e. a Kolmogorov consistent family, which is

the input of an extension problem where the designs play the role of the finite sets of indices. The corresponding universal property is the one of the Kolmogorov products of [24, Def. 3.1, Def. 4.1], which exist in **BorelStoch** by the classical extension theorem [24, Ex. 3.6]. Coherence over Ω_{red} gives no symmetry, since between two objects there is at most one inclusion. Relabelling-coherence, i.e. coherence over Ω_{inj} , gives in addition, at the automorphisms of each $\omega_n = \{1, \dots, n\}$, the invariance under the finite permutations which the categorical de Finetti theorem [11, Thm. 4.4] assumes for a state on a countable Kolmogorov power, in a Markov category which satisfies [11, Assumption 4.2], as **BorelStoch** does [11, Ex. 4.3]. Hence exchangeability is neither an additional modelling assumption nor a consequence of the design reductions alone. It comes from the relabellings, which are the morphisms that distinguish the two classical criteria. This is only a structural observation. We do not prove an extension theorem for the setting of McCullagh and Brøns, and we have not checked the hypotheses of [24, 11] against Definition 2.1.

What is computed

The script `oneway_prior_coherence.py` carries out the computations of Propositions 8.4 and 8.5 in exact arithmetic over $\mathbb{Q}$. It is deposited with its committed output in the same release of <https://github.com/Vaxx66/calamus> as the scripts of Section 6, and a fresh run reproduces the committed output byte for byte. For each of the four classes of Proposition 8.4, the script builds the linear system in the unknown $(\pi_{\omega_1}, \pi_{\omega_2})$ given by the identities (19) along the generators of the class and by the two normalisations, it computes the affine dimension of the set of the solutions, and it compares it with the formulas for $|M| \leq 4$ and $|S| \leq 3$. Then it evaluates the analogue of Jeffreys' rule and the product rules on some cases, and it computes exactly the total variation defects along i_1, i_2 and p , including the case where the bound $1 - |M|^{-1}$ for the merge is strict.

9 Ridge regression as a bridged Bayesian object

In this section we apply the constructions of Section 8 to the Gaussian linear model of Example 2.12, and we prove part (3) of Theorem 1.7. We obtain the ridge estimator, together with the correspondence between the penalty and the precision of the prior. We shall see that the ridge prior is coherent in the sense of Definition 8.1, and that its failure of naturality over the whole category of the covariate spaces corresponds to the usual requirement that ridge regression be applied to standardised covariates.

Throughout this section $\sigma^2 = \sigma_0^2$ is fixed and known, and the unknown parameter is the coefficient functional. All the objects below, i.e. $\omega^* \cong \mathbb{R}^p$, $\mathbb{R}_{>0}$ and $\mathbb{R}^n \cong \mathbb{R}^n$, are standard Borel. Hence the convention of Section 7 applies, and the Bayesian inversions exist and are unique up to a.s. equality.

Definition 9.1 (Ridge datum). A *ridge datum* on Ω is a choice, for each covariate space ω , of an inner product $\langle \cdot, \cdot \rangle_\omega$ on ω , equivalently of a basis up to orthogonal transformations, together with a real number $\tau^2 > 0$ which does not depend on ω . The datum is called *compatible* if the inner products restrict along the inclusions of Ω_{red} , i.e. if $\langle \cdot, \cdot \rangle_{\omega'} = \langle \cdot, \cdot \rangle_\omega \upharpoonright_{\omega'}$ whenever $\omega' \subseteq \omega$, so that every insertion is an isometric embedding. Let I_ω be the form induced on ω^* . The *ridge prior* associated to the datum is the product state

$$\pi_\omega^\tau := \mathcal{N}(0, \tau^2 I_\omega) \otimes \delta_{\sigma_0^2} : I \rightsquigarrow \omega^* \times \mathbb{R}_{>0}.$$

A Gaussian state on ω^* is not canonical, since it requires the choice of a positive definite quadratic form, and this is the reason for the choice of the inner products. Compatible data exist on the full subcategory of $\mathbf{Vect}_{\mathbb{R}}^{\text{fd}}$ whose objects are the linear subspaces of the coordinate spaces $\mathbb{R}^p$, $p \geq 0$, which contains a representative of every finite dimensional space and all the linear maps between its objects. Indeed, take the standard form on each $\mathbb{R}^p$ and its restriction on each subspace, which is the choice implicit in the description by matrices of Example 2.12(5). In this section Ω denotes this subcategory. Compatibility is a condition on the datum, and it is used only in Proposition 9.4(1).

Proposition 9.2 (Ridge as Bayesian inversion). *Let $d = (u, \omega, x)$ be a design. Choose a basis of ω which is orthonormal for the ridge datum and an enumeration of u , so that x becomes the $n \times p$ matrix X of*

Example 2.12(5), and set

$$\lambda := \frac{\sigma_0^2}{\tau^2}.$$

Let $\mathbf{P}_d$ be the component of the bridged model of the Gaussian linear model (Theorem 4.4).

1. The prior predictive state is $m_d = \mathcal{N}_n(0, \tau^2 XX^\top + \sigma_0^2 I_n)$.
2. The Bayesian inversion $\mathbf{P}_d^\dagger: \mathbb{R}^u \rightsquigarrow \omega^* \times \mathbb{R}_{>0}$ exists in $\mathbf{BorelStoch}$, and it is the Gaussian kernel

$$\mathbf{P}_d^\dagger(y) = \mathcal{N}_p((X^\top X + \lambda I)^{-1} X^\top y, \sigma_0^2 (X^\top X + \lambda I)^{-1})$$

tensored with $\delta_{\sigma_0^2}$. We omit the factor $\delta_{\sigma_0^2}$ here and in what follows.

3. The mean of the posterior is the ridge estimator of Hoerl and Kennard [15, eq. (2.1), p. 57], $\widehat{\beta}_\lambda(y) = (X^\top X + \lambda I)^{-1} X^\top y$, and the penalty is the ratio of the two variances, i.e. $\lambda = \sigma_0^2 / \tau^2$. This is the Bayesian reading which is already given in [15, §6, p. 64], where the ridge estimate is the posterior mean under a centred normal prior with covariance $(\delta^2/k)I$, where δ^2 is not defined there and must be the error variance σ^2 for the statement to hold, so that $k = \sigma^2 / \tau^2$.

Proof. Let β be the coordinate vector of the coefficient functional. Under π_ω^τ and $\mathbf{P}_d$ we have $\beta \sim \mathcal{N}_p(0, \tau^2 I)$ and $y \mid \beta \sim \mathcal{N}_n(X\beta, \sigma_0^2 I)$. Hence (β, y) is jointly Gaussian with $\text{Cov}(y) = \tau^2 XX^\top + \sigma_0^2 I$ and $\text{Cov}(\beta, y) = \tau^2 X^\top$, and this gives (1). The precision of the posterior is $\sigma_0^{-2} X^\top X + \tau^{-2} I = \sigma_0^{-2} (X^\top X + \lambda I)$, which gives the stated covariance, and the mean of the posterior is that covariance applied to $\sigma_0^{-2} X^\top y$, i.e. $(X^\top X + \lambda I)^{-1} X^\top y$. Since $X^\top X + \lambda I$ is positive definite for $\lambda > 0$, these expressions are defined with no condition on the rank of X . Finally, this Gaussian kernel is the Bayesian inversion because $(\mathbf{P}_d^\dagger \otimes \text{id}) \circ \text{copy} \circ m_d$ is the joint state of Definition 8.1, which for jointly Gaussian pairs is the classical formula for conditioning. This gives (2), and (3) follows. $\square$

Remark 9.3. The content of Proposition 9.2 is the classical conjugacy. The categorical statements are that π_ω^τ is a state, that $\mathbf{P}_d$ is a morphism of $\mathbf{Stoch}$ (Theorem 4.4), and that the inversion exists in $\mathbf{BorelStoch}$. Thus, prior, likelihood, joint distribution and posterior are four morphisms of the same category, and they can be composed with the structure indexed by the designs of Section 4.

Proposition 9.4 (Coherence of the ridge prior). *Let π^τ be as in Definition 9.1, with Ω as fixed after that definition, and let $\psi: \omega \rightarrow \omega'$ be a morphism of Ω , so that $\Theta(\psi) = \psi^* \times \text{id}$.*

1. *If ψ is an isometric embedding for the chosen inner products, then $\Theta(\psi)_* \pi_{\omega'}^\tau = \pi_\omega^\tau$. Hence, for a compatible ridge datum, π^τ is coherent, i.e. Ω_{red} -coherent in the sense of Definition 8.1, and by Proposition 8.2(3) the family of the prior predictives $(m_d)_d$ is natural over Ω_{red} .*
2. *Let A be a linear automorphism of ω . Then (19) holds at A if and only if A is orthogonal for the chosen inner product. In particular it fails for every rescaling of the coordinates $\text{diag}(a_1, \dots, a_p)$ with some $|a_j| \neq 1$. Hence π^τ is never Ω_{inj} -coherent when some ω has $\dim \omega \geq 1$, whatever the datum is.*
3. *Let ψ be surjective with $\dim \omega' < \dim \omega$, i.e. a merging or a collapsing of covariate directions. Then ψ^* has a proper image, and by Proposition 8.2(1) coherence at ψ would confine π_ω^τ to a proper subspace, which is not possible for a nondegenerate Gaussian.*
4. *Suppose that X has full column rank. Then $\alpha \mapsto \mathbf{P}_d \circ (\alpha \otimes \delta_{\sigma_0^2})$ is injective on the states α of ω^* , i.e. the hypothesis of Proposition 8.2(4) holds at d on the class of the states supported on $\omega^* \times \{\sigma_0^2\}$, which contains the ridge priors and is preserved by the maps $\Theta(\psi)_*$. Hence, among the priors with variance fixed at σ_0^2 , the naturality of the prior predictives over Ω_{red} is equivalent to the coherence of the prior.*

Proof. (1) Choose an orthonormal basis of ω for $\langle, \rangle_\omega$ and extend it to an orthonormal basis of ω' , which is possible because ψ is an isometric embedding. In the dual coordinates ψ^* is the coordinate projection $\mathbb{R}^{p'} \rightarrow \mathbb{R}^p$, and the pushforward of $\mathcal{N}(0, \tau^2 I_{p'})$ along a coordinate projection is $\mathcal{N}(0, \tau^2 I_p)$. The factor $\delta_{\sigma_0^2}$ is preserved by id. For a compatible datum every morphism of Ω_{red} is an isometric embedding.

(2) The map $\Theta(A)$ acts on the coordinate vectors as $A^\top$, hence $\Theta(A)_* \mathcal{N}(0, \tau^2 I) = \mathcal{N}(0, \tau^2 A^\top A)$, which is equal to $\mathcal{N}(0, \tau^2 I)$ if and only if $A^\top A = I$.

(3) It follows from Proposition 8.2(1) applied to $\text{im } \psi^* = (\ker \psi)^0$, the annihilator of $\ker \psi$.

(4) Under $\alpha \otimes \delta_{\sigma_0^2}$ the prior predictive is the location mixture $\int \mathcal{N}_n(X\beta, \sigma_0^2 I) \alpha(d\beta)$, whose characteristic function is $t \mapsto e^{-\sigma_0^2 |t|^2/2} \widehat{X_* \alpha}(t)$. It determines $X_* \alpha$, hence α , because X is injective. The restriction to this class cannot be dropped. For $n = p$ the states $\mathcal{N}(0, c(X^\top X)^{-1}) \otimes \delta_{\sigma_0^2 - c}$, with $0 < c < \sigma_0^2$, and $\delta_0 \otimes \delta_{\sigma_0^2}$ have the same prior predictive $\mathcal{N}(0, \sigma_0^2 I)$. $\square$

Proof of Theorem 1.7. Parts (1) and (2) are Proposition 8.2(1) and (2). Part (3) follows from Proposition 9.2 and Proposition 9.4(1) and (2). $\square$

Remark 9.5. The ridge prior carries a choice of geometry on the covariates, and by Proposition 9.4(2) this choice is not preserved by the recodings. Thus, in terms of the classes of Section 5, the prior is coherent over the class of Tjur and not over the class of McCullagh. This is the categorical form of two standard facts. The first one is the instruction to apply ridge regression to standardised covariates, which is already present in the original paper, where the model is formulated so that $X^\top X$ is in correlation form, and ill-conditioning is measured by its distance from the identity in that form [15, pp. 55–56]. The second one is that $\widehat{\beta}_\lambda$, unlike the least squares estimator, is not equivariant under the linear reparametrisations of the covariate space. Note that the model $\mathbf{P}$ is natural over Ω (Lemma 2.13), so that the failure of naturality comes from the prior only.

A The category of statistical models and the arrows of Stoch

In this appendix we compare the category StatMod of Brøns (Definition 2.4) with the category whose objects are the morphisms of Stoch and whose morphisms are the commuting squares with deterministic sides.

A.1 Copowers and squares

The monoidal unit I of Stoch is the one-point measurable space, and

$$\text{Stoch}(I, S) = \text{Meas}(1, \mathcal{G}S) = \mathcal{G}S = \mathcal{P}(S)$$

naturally in S , i.e. $\mathcal{P} \cong \text{Stoch}(I, -)$. The Dirac embedding δ (Definition 3.3) is the free functor of the Kleisli adjunction $\text{Meas} \rightleftarrows \text{Stoch}$, hence it is a left adjoint, and it preserves colimits. The coproducts in Meas are the disjoint unions, with the disjoint-union σ -algebra. Hence for every set A the discrete space $(A, 2^A)$ is the copower of I in Stoch, i.e.

$$\delta(A, 2^A) \cong \coprod_A I =: A \cdot I.$$

Therefore, for every set A and every measurable space S ,

$$\text{Set}(A, \mathcal{P}(S)) \cong \text{Set}(A, \text{Stoch}(I, S)) \cong \text{Stoch}(A \cdot I, S), \quad (20)$$

which is the instance of (5) used in the proof of Theorem 4.1. For $p: A \rightarrow \mathcal{P}(S)$ we denote by $\widehat{p}: A \cdot I \rightsquigarrow S$ the corresponding kernel $\widehat{p}(a)(B) = p(a)(B)$, which is measurable in a because $A \cdot I$ is discrete. For a kernel $q: A \cdot I \rightsquigarrow S$ we denote by $\check{q}: a \mapsto q(a)$ the corresponding map.

Definition A.1. Let $q: X \rightsquigarrow Y$ and $q': X' \rightsquigarrow Y'$ be morphisms of **Stoch**. A *square* from q to q' is a pair (a, b) of deterministic morphisms $a: X \rightsquigarrow X'$ and $b: Y \rightsquigarrow Y'$, called the *legs* of the square, such that $q' \circ a = b \circ q$. The squares compose componentwise. We denote by $\text{Stoch}_{\text{det}}^{\rightarrow}$ the category whose objects are the morphisms of **Stoch** and whose morphisms are the squares, and by $\text{Stoch}_{\text{det}, \text{cop}}^{\rightarrow}$ its full subcategory of the objects $q: X \rightsquigarrow Y$ such that X is isomorphic in **Stoch** to a copower $A \cdot I$.

Lemma A.2. Let $f: X \rightarrow X'$ and $g: Y \rightarrow Y'$ be measurable maps. Then $(\delta f, \delta g)$ is a square from q to q' if and only if $q'(f(x)) = g_* q(x)$ in $\mathcal{P}(Y')$ for every $x \in X$.

Proof. We have $(q' \circ \delta f)(x) = q'(f(x))$ by (9) and $(\delta g \circ q)(x) = g_* q(x)$ by (8). $\square$

We shall use the following two conditions.

- (a) For a set A : every countably additive $\{0, 1\}$ -valued probability measure on $(A, 2^A)$ is a Dirac measure.
- (b) For a measurable space Z : the σ -algebra Σ_Z is countably generated and separates points. By Lemma 3.4(2) this implies that for every W the deterministic morphisms $W \rightsquigarrow Z$ are the Dirac kernels of the measurable maps.

Lemma A.3 (Ulam). *We call a countably additive $\{0, 1\}$ -valued probability measure on $(A, 2^A)$ which is not a Dirac measure a two-valued measure on the set A .*

1. *The two-valued measures on A correspond bijectively to the countably complete non-principal ultrafilters on A [16, (10.3)–(10.4)].*
2. *If A carries no two-valued measure, then neither does any set of cardinality at most $|A|$, nor the power set 2^A [27, Bem. 2 and Satz 1]. In particular, no set of cardinality at most that of the continuum carries one.*
3. *The set A carries a two-valued measure if and only if there is a measurable cardinal $\kappa \leq |A|$ [16, Lemma 10.2 and Def. 10.3]. The original form of Ulam is that A carries none when no weakly inaccessible cardinal is $\leq |A|$ [27, Satz (A)], cf. [16, Thm. 10.1].*

Thus condition (a) says that no measurable cardinal is $\leq |A|$. By Lemma A.3(2), it holds, with no set-theoretic hypothesis, when A has at most the cardinality of the continuum, which is the case for all the parameter sets of this paper.

A.2 The comparison functor

Proposition A.4. *The bijection (20) extends to a functor*

$$\Delta: \text{StatMod} \longrightarrow \text{Stoch}_{\text{det}}^{\rightarrow}, \quad (\Theta_0, S, p) \longmapsto (\widehat{p}: \Theta_0 \cdot I \rightsquigarrow S), \quad (r, f) \longmapsto (\delta r, \delta f),$$

with the following properties.

1. *Δ is injective on the objects, and it gives a bijection between the objects of **StatMod** and the arrows of **Stoch** whose domain is a copower $A \cdot I$. Hence Δ is essentially surjective onto $\text{Stoch}_{\text{det}, \text{cop}}^{\rightarrow}$.*
2. *Let $x = (\Theta_0, S, p)$ and $x' = (\Theta'_0, S', p')$ be objects of **StatMod**. If Θ'_0 satisfies (a) and S' satisfies (b), then the map $\text{StatMod}(x, x') \rightarrow \text{Stoch}_{\text{det}}^{\rightarrow}(\Delta x, \Delta x')$ induced by Δ is a bijection.*
3. *Let $\text{StatMod}_{(a),(b)} \subseteq \text{StatMod}$ be the full subcategory of the objects (Θ_0, S, p) such that Θ_0 satisfies (a) and S satisfies (b). Then Δ restricts to an equivalence between $\text{StatMod}_{(a),(b)}$ and the full subcategory of $\text{Stoch}_{\text{det}, \text{cop}}^{\rightarrow}$ of the arrows $X \rightsquigarrow S$ such that S satisfies (b) and $X \cong A \cdot I$ with A satisfying (a). Let $\text{StatMod}_{\text{fin}}$ be the full subcategory of the objects with Θ_0 finite and S finite and discrete. Then Δ gives an isomorphism of categories $\text{StatMod}_{\text{fin}} \cong \text{FinStoch}_{\text{det}}^{\rightarrow}$.*

Proof. Let $(r, f): (\Theta_0, S, p) \rightarrow (\Theta'_0, S', p')$ be a morphism of **StatMod**, i.e. $\mathcal{P}(f) \circ p = p' \circ r$ (Definition 2.4). The map r is measurable because its domain is discrete, hence δr and δf are Dirac kernels, and they are deterministic. By Lemma A.2 the pair $(\delta r, \delta f)$ is a square from $\widehat{p}$ to $\widehat{p}'$ if and only if $\widehat{p}'(r(\theta)) = f_* \widehat{p}(\theta)$ for every θ , i.e. if and only if $p' \circ r = \mathcal{P}(f) \circ p$. Since δ is a functor and the squares compose componentwise, Δ is a functor.

(1) The assignment $(\Theta_0, S, p) \mapsto \widehat{p}$ is a bijection from the objects of **StatMod** onto the arrows of **Stoch** whose domain is of the form $A \cdot I = (A, 2^A)$, with inverse $q \mapsto (A, S, \check{q})$. Indeed, the domain of $\widehat{p}$ gives Θ_0 as its underlying set, the codomain gives S , and $p = \check{\widehat{p}}$. Hence Δ is injective on the objects and its image is contained in $\text{Stoch}_{\text{det, cop}}^{\rightarrow}$. Let now $q: X \rightsquigarrow S$ be an object of $\text{Stoch}_{\text{det, cop}}^{\rightarrow}$, and let $v: A \cdot I \rightsquigarrow X$ and $u: X \rightsquigarrow A \cdot I$ be mutually inverse. We show that they are deterministic. For $a \in A$ let $X_a := \{x \in X : u(x)(\{a\}) = 1\}$, which is measurable. Since $u \circ v = \text{id}_{A \cdot I}$ we have

$$1 = \delta_a(\{a\}) = (u \circ v)(a)(\{a\}) = \int_X u(x)(\{a\}) v(a)(dx),$$

and the integrand is bounded by 1, hence $v(a)(X_a) = 1$. In particular $X_a \neq \emptyset$. For $x \in X_a$ the probability measure $u(x)$ on 2^A gives mass 1 to $\{a\}$, i.e. $u(x) = \delta_a$. Since $v \circ u = \text{id}_X$ we obtain

$$\delta_x(F) = (v \circ u)(x)(F) = \int_A v(b)(F) u(x)(db) = v(a)(F) \quad (F \in \Sigma_X).$$

Thus $v(a) = \delta_x$ for every $x \in X_a$. Choosing $x_a \in X_a$ for each a we obtain a map $f: A \rightarrow X$, which is measurable because its domain is discrete, such that $v = \delta f$. Hence v is deterministic, and u is deterministic by [10, Lem. 10.9(c)]. Now $q \circ v: A \cdot I \rightsquigarrow S$ has domain $A \cdot I$, so that $q \circ v = \Delta(A, S, (q \circ v)^\vee)$. Furthermore $(v, \text{id}_S): q \circ v \rightarrow q$ and $(u, \text{id}_S): q \rightarrow q \circ v$ are squares, and they are mutually inverse in $\text{Stoch}_{\text{det}}^{\rightarrow}$. Hence $q \cong \Delta(A, S, (q \circ v)^\vee)$.

(2) Let (a, b) be a square from $\widehat{p}$ to $\widehat{p}'$. The leg $a: \Theta_0 \cdot I \rightsquigarrow \Theta'_0 \cdot I$ is deterministic, hence each $a(\theta)$ is a countably additive $\{0, 1\}$ -valued probability measure on $2^{\Theta'_0}$ [10, Ex. 10.4]. By (a) it is $\delta_{r(\theta)}$ for a unique $r(\theta) \in \Theta'_0$, and r is measurable because Θ_0 is discrete, so that $a = \delta r$. By (b) for S' we have $b = \delta f$ for a unique measurable map $f: S \rightarrow S'$. By Lemma A.2 we have $p' \circ r = \mathcal{P}(f) \circ p$, i.e. (r, f) is a morphism of **StatMod** with $\Delta(r, f) = (a, b)$. Hence the map $\text{StatMod}(x, x') \rightarrow \text{Stoch}_{\text{det}}^{\rightarrow}(\Delta x, \Delta x')$ is surjective. It is injective by Lemma 3.4(1), since $2^{\Theta'_0}$ separates points and $\Sigma_{S'}$ does by (b).

(3) Let $\mathcal{A}$ be the full subcategory of $\text{Stoch}_{\text{det, cop}}^{\rightarrow}$ of the statement. The functor Δ restricts to a functor $\text{StatMod}_{(a), (b)} \rightarrow \mathcal{A}$, which is fully faithful by part (2). It is essentially surjective by the argument of part (1). Indeed, for $q: X \rightsquigarrow S$ in $\mathcal{A}$ with $v: A \cdot I \rightsquigarrow X$ an isomorphism and A satisfying (a), the object $(A, S, (q \circ v)^\vee)$ belongs to $\text{StatMod}_{(a), (b)}$ and $q \cong \Delta(A, S, (q \circ v)^\vee)$. Hence it is an equivalence [20, IV.4, Thm. 1(iii)]. Finally every object of **FinStoch** is $A \cdot I$ for a finite set A , so that by part (1) Δ gives a bijection from the objects of $\text{StatMod}_{\text{fin}}$ onto the morphisms of **FinStoch**, i.e. onto the objects of $\text{FinStoch}_{\text{det}}^{\rightarrow}$. Conditions (a) and (b) hold for every finite set and every finite discrete space, hence Δ is bijective on the hom-sets by part (2), and a functor which is bijective on the objects and on the hom-sets is an isomorphism of categories. $\square$

A.3 The measurable analogue

The restriction to the arrows with copower domain comes from the **Set**-valued formulation. Consider the comma category

$$\text{StatMod}_{\text{Meas}} := (\text{id}_{\text{Meas}} \downarrow \mathcal{G}).$$

Its objects are the measurable maps $\Theta \rightarrow \mathcal{G}S$, i.e. by (5) the morphisms $\Theta \rightsquigarrow S$ of **Stoch**, with no condition on the domain. We denote by $D: \text{Set} \rightarrow \text{Meas}$ the functor which endows a set with the σ -algebra of all its subsets and by $U: \text{Meas} \rightarrow \text{Set}$ the underlying-set functor.

Proposition A.5. *We have $D \dashv U$ and $U\mathcal{G} = \mathcal{P}$. Let $\varepsilon: DU \Rightarrow \text{id}_{\text{Meas}}$ be the counit, whose component ε_X is the identity map of UX , and let $\lambda := \varepsilon\mathcal{G}: D\mathcal{P} \Rightarrow \mathcal{G}$.*

1. The assignments $(\Theta_0, S, p) \mapsto (D\Theta_0, S, \lambda_S \circ Dp)$ and $(r, f) \mapsto (Dr, f)$ give a functor $\Phi: \text{StatMod} \rightarrow \text{StatMod}_{\text{Meas}}$.
2. The assignments $(\Theta, S, q) \mapsto (U\Theta, S, Uq)$ and $(a, f) \mapsto (Ua, f)$ give a functor $\Psi: \text{StatMod}_{\text{Meas}} \rightarrow \text{StatMod}$, and $\Phi \dashv \Psi$.
3. We have $\Psi\Phi = \text{id}_{\text{StatMod}}$. Hence Φ is fully faithful and StatMod is a coreflective subcategory of $\text{StatMod}_{\text{Meas}}$ with coreflector Ψ .
4. The counit of $\Phi \dashv \Psi$ at (Θ, S, q) is $(\varepsilon_\Theta, \text{id}_S)$, which is invertible if and only if $\Sigma_\Theta = 2^\Theta$. The essential image of Φ is the full subcategory of $\text{StatMod}_{\text{Meas}}$ of the objects whose parameter space is discrete.

No hypothesis on the spaces is needed.

Proof. We have $\text{Meas}(DA, X) = \text{Set}(A, UX)$ naturally, because every map from a discrete space is measurable, hence $D \dashv U$. The underlying set of $\mathcal{GS}$ is $\mathcal{P}(S)$ and the underlying map of $\mathcal{G}(f)$ is the push-forward, hence $U\mathcal{G} = \mathcal{P}$. The transformation λ is natural, and each λ_S is the identity on the underlying sets.

(1) The map $\lambda_S \circ Dp$ is measurable, hence it is an object of $\text{StatMod}_{\text{Meas}}$. Let (r, f) be a morphism of StatMod , i.e. $\mathcal{P}(f) \circ p = p' \circ r$. By the naturality of λ we have

$$\mathcal{G}(f) \circ \lambda_S \circ Dp = \lambda_{S'} \circ D(\mathcal{P}(f)) \circ Dp = \lambda_{S'} \circ D(p' \circ r) = (\lambda_{S'} \circ Dp') \circ Dr,$$

i.e. (Dr, f) is a morphism of $\text{StatMod}_{\text{Meas}}$. Identities and compositions are preserved componentwise.

(2) The functor Ψ is well defined since $Uq: U\Theta \rightarrow U\mathcal{GS} = \mathcal{P}(S)$. A morphism $\Phi(\Theta_0, S, p) \rightarrow (\Theta', T, q)$ is a pair (a, f) , where $a: D\Theta_0 \rightarrow \Theta'$ is measurable and $q \circ a = \mathcal{G}(f) \circ \lambda_S \circ Dp$. Let $r = Ua: \Theta_0 \rightarrow U\Theta'$ be the map which corresponds to a under $D \dashv U$. Both sides of the equality are measurable maps from the discrete space $D\Theta_0$ and U is faithful, hence the equality is equivalent to $Uq \circ r = \mathcal{P}(f) \circ p$, i.e. to the fact that (r, f) is a morphism $(\Theta_0, S, p) \rightarrow \Psi(\Theta', T, q)$ of StatMod . This correspondence is bijective and natural in both variables.

(3) We have $UD = \text{id}_{\text{Set}}$ and $U\lambda_S = \text{id}$, hence $\Psi\Phi$ is the identity on the objects and on the morphisms. A left adjoint whose unit is invertible is fully faithful, and its domain is then coreflective in its codomain.

(4) We have $\Phi\Psi(\Theta, S, q) = (DU\Theta, S, \lambda_S \circ D(Uq))$, and $(\varepsilon_\Theta, \text{id}_S)$ is a morphism from it to (Θ, S, q) , since both sides of the required equality are measurable maps $DU\Theta \rightarrow \mathcal{GS}$ with underlying map Uq . The triangle identities show it is the counit. The bijection ε_Θ is an isomorphism of Meas if and only if $\Sigma_\Theta = 2^\Theta$, and $X \cong DA$ in Meas implies $\Sigma_X = 2^X$. This gives the essential image. $\square$

Remark A.6. A morphism (a, f) of $\text{StatMod}_{\text{Meas}}$ is a pair of measurable maps, and $(a, f) \mapsto (\delta a, \delta f)$ gives a functor $\Delta_{\text{Meas}}: \text{StatMod}_{\text{Meas}} \rightarrow \text{Stoch}_{\text{det}}^\rightarrow$ by Lemma A.2. It is bijective on the objects by (5). By Lemma 3.4(1) and (2) it is bijective on the morphisms into an object (Θ', S', q') when Θ' and S' satisfy (b). Hence Δ_{Meas} restricts to an equivalence between the full subcategory of $\text{StatMod}_{\text{Meas}}$ of the objects whose parameter space and sample space satisfy (b) and the full subcategory of $\text{Stoch}_{\text{det}}^\rightarrow$ of the arrows between such spaces. We have $\Delta \cong \Delta_{\text{Meas}}\Phi$, where the isomorphism is the identification $D\Theta_0 \cong \Theta_0 \cdot I$ in Stoch . The functor Φ needs no hypothesis, while conditions (a) and (b) are used only for Δ_{Meas} and Δ . We do not know whether they are necessary. Note that there is no equivalence with the full arrow category $\text{Stoch}^\rightarrow$, whose squares have arbitrary kernels as legs, because the squares which come from pairs of measurable maps are deterministic.

Corollary A.7. Let $\mathbf{C}$ be a small category. The composition with Φ and Ψ gives functors Φ_* and Ψ_* between the categories of diagrams $[\mathbf{C}, \text{StatMod}]$ and $[\mathbf{C}, \text{StatMod}_{\text{Meas}}]$ such that $\Phi_* \dashv \Psi_*$ and $\Psi_*\Phi_* = \text{id}$. For $\mathbf{C} = \mathbf{V} \times \mathbf{D}^{\text{op}}$, the bijection of Proposition 2.5 gives a bijection between the McCullagh–Brøns models and the functors $\mathbf{V} \times \mathbf{D}^{\text{op}} \rightarrow \text{StatMod}_{\text{Meas}}$ whose two projections are $D \circ L$ and $\Gamma \circ (\text{id}_{\mathbf{V}} \times \text{pr}_{\mathbf{D}}^{\text{op}})$. A diagram in $\text{StatMod}_{\text{Meas}}$ belongs to the essential image of Φ_* if and only if its parameter spaces are discrete.

Proof. The composition with an adjunction is an adjunction between the categories of diagrams, with unit and counit computed pointwise, and $\Psi_*\Phi_* = (\Psi\Phi)_* = \text{id}$. A functor $F: \mathbf{V} \times \mathbf{D}^{\text{op}} \rightarrow \text{StatMod}_{\text{Meas}}$ with the stated projections has parameter spaces of the form $D\Theta_0$, hence $F = \Phi_*\Psi_*F$, since both sides have the same underlying maps and every map from a discrete space is measurable. Therefore $F \mapsto \Psi_*F$ and $G \mapsto \Phi_*G$ are mutually inverse bijections between these functors and the functors $\mathbf{V} \times \mathbf{D}^{\text{op}} \rightarrow \text{StatMod}$ of Proposition 2.5(2). The last statement follows from Proposition A.5(4), since the counit is computed pointwise and discreteness is invariant under the isomorphisms of Meas . $\square$

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